Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Black Sea Production Centre
BLKSEA_MULTIYEAR_WAV_007_006

Issue: 4.1

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table.

The third column specifies which sections or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

Issue	Date	§	Description of Change	Author	Validated By
1.0	01/09/2019	All	First version of document of CMEMS V4	J. Staneva, A. Behrens, G. Gayer, O. Krüger	E. Peneva
2.0	07/12/2020	All	Complete revision due to new model version	J. Staneva, A. Behrens, M. Ricker, G. Gayer	E. Peneva
2.1	15/01/2021	§I, §VI	Addition of the interim dataset	J. Staneva, A. Behrens	E. Peneva
2.2	10/09/2021	§VI	Addition of the 2020 time series extension	J. Staneva, M. Ricker, A. Behrens	E. Peneva
3.0	29/11/2022	All	Complete revision due to new model version	J. Staneva, M. Ricker, A. Behrens	E. Peneva
3.1	30/11/2023	All	Addition of time series extension to 1950 and addition of LATEMAR max. variables	J. Staneva, M. Ricker, A. Behrens	E. Peneva
3.2	18/06/2024	I.1, I.2, III, VI	Removal of the cold start period (first week of 1950)	M. Ricker	E. Peneva
4.0	26/11/2024	All	Addition of PHY forcing (sea surface height and surface currents) and of a new air-sea-flux dataset	J. Staneva, M. Ricker, A. Behrens, N. T. Pham	E. Peneva
4.1	28/05/2025	All	Merging of the Interim and Reanalysis datasets, Adaption to latest template	M. Ricker	E. Peneva, R.Lecci

ACRONYM TABLE

Acronym	Description
ADCP	Acoustic Doppler Current Profiler
AVISO	Archiving, Validation and Interpretation of Satellite Oceanographic data (service of French space agency CNES)
BLK	Black Sea
EAN	Estimated Accuracy Number
ECMWF	European Centre for Medium-Range Weather Forecasts
EIS	Entry Into Service
Hmax	Maximum Wave Height
JS1/2/3	Jason-1/2/3 Satellite Missions
M	Month
MFC	Monitoring Forecasting Centre
MY	Multi-year
MYP	Multi-Year Product
NRT	Near Real Time
OI	Optimal Interpolation
SWH	Significant Wave Height (VHM0)
TAC	Copernicus Marine Service Thematic Assembly Centre
T_{M02}	T_{M02} Mean Wave Period
Tp	Wave Peak Period
U10	Wind speed at 10 metre height
WAM	WAve Model (numerical wave model)
WAV	Waves
Y	Year

KEY DEFINITIONS

Numerical wave model	Numerical model based on gravity wave spectra solving the energy density balance equation.
Significant wave height	Average height of the highest one-third of the waves (computed from spectra by: $SWH = 4\sqrt{m_0}$ with m_0 = zeroth spectral moment).
Swell	Waves not correlated with the local wind field (wave phase speed > wind speed).
Wave spectrum	Wave energy at each wave frequency for each direction.
Wind sea	Waves generated and characterised by local winds (wave phase speed < wind speed).

RELEVANT DOCUMENT LIST

[DA1] CMEMS-BLK-PUM-007-006	Black Sea Waves Reanalysis Product User Manual: https://documentation.marine.copernicus.eu/PUM/CMEMS-BLK-PUM-007-006.pdf
[DA2] CMEMS-BLK-SQO-007-006	Black Sea Waves Reanalysis Synthesis Quality Overview Document: https://documentation.marine.copernicus.eu/SQO/CMEMS-BLK-SQO-007-006.pdf

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of the new Multi-Year Product (MYP) for the Black Sea (BLK) wave (WAV) component (BLKSEA_MULTIYEAR_WAV_007_006). The BLK-WAV product includes the complete hindcast time period 08/01/1950 to -18M (i.e., 18 months before present). The product includes the 2D 1-hourly analysis and forecast instantaneous fields (list available in Section II.1). The output data are produced at 1/40°x1/40° horizontal resolution.

Interim files are also provided: they consist of operational datasets that will be updated every month (M) to cover the month M-1 and are produced by the BLK-WAV MY system, forced by ECMWF (European Centre for Medium-Range Weather Forecasts) ERA5 atmospheric fields. Details on the processing system are provided in [DA1].

I.2 Summary of the results

The quality of the MY product BLKSEA_MULTIYEAR_WAV_007_006 has been assessed via comparing the reanalyses against satellite observations recorded by the radar altimeters of Jason-1, Jason-2 and Jason-3, for the time period from 01/01/2002 to 31/12/2021, independent non-assimilated satellite data, and available in-situ observations. For the time period 08/01/1950 to 31/12/2001 no satellite measurements were available for the comparison. The horizontal spatial grid resolution of the BLK-WAV model is 1/40° in both zonal and meridional directions (ca. 2.5 km).

The main results of the BLKSEA_MULTIYEAR_WAV_007_006 quality product assessment are summarized below:

Significant wave height: The comparisons of the significant wave height (SWH) have been done with satellite altimeter data (assimilated and non-assimilated data) for the period from 2002 to 2021 and in-situ observations. The V4 BLK-WAV system used to produce the product BLKSEA_MULTIYEAR_WAV_007_006 presents good accuracy in terms of SWH. The model skill enhancement compared to the old product, based on different statistical parameters, is evident and mainly related to a reduced bias and reduced differences between model and satellite data. The skill critically depends upon the quality and the temporal and spatial resolution of the wind forcing for the Black Sea. The newly introduced wind assessments reveal good wind data quality with underestimated high wind speeds. However, SWHs show no distinct deviation from the measurements over the full range of wave heights. Overall, with respect to assimilated SWH satellite data, the wave model results and observations are correlated at a level of 0.94. In general, the wave model tends to slightly overestimate the satellite measurements. The bias (mean of the difference: model minus measurements) is small with a value of 0.02 m. Using non-assimilated data, the metrics are similar with values of 0.93 and -0.04 m, respectively. Furthermore, the BLKSEA_MULTIYEAR_WAV_007_006 SWH product captures the temporal variability very well for almost all parts of the distribution of SWHs.

Wave Buoy validations: For the period May 2020 to December 2021, time series of eleven buoys (SPOT0772, SPOT0773, SPOT0776, WAVEB01, WAVEB02, WAVEB03, WAVEB04, WAVEB05, WAVEB06, WD3044, Gelendzhik) were available (INSITU_BLK_PHYBGCWAV_DISCRETE_MYNRT_013_034). The water depth at the buoy locations is often shallow (17 m or 20 m) and the wave buoys are located close to the coast. The quality control of some of the stations is still insufficient, which limits the validations

against in-situ observations. Nevertheless, the modelled SWH shows a good correlation with buoy data (0.91). In addition to SWH, the in-situ observations provide other wave parameters, such as maximum wave height, mean direction, peak direction, T_{M02} period, and peak period. The correlation between the model and buoy data for these parameters is also very good (Hmax: 0.83, T_{M02}: 0.67, and Tp: 0.67). Some SWH peaks are slightly underestimated by the model, likely due to the buoys' proximity to the coast, a too coarse model resolution, or discrepancies between the model and real bathymetry. In the future, buoy data for past periods will be collected and integrated into the quality control in order to assess past periods of the dataset.

I.3 Estimated Accuracy Numbers

Estimated accuracy numbers (EANs) for the results of the BLK-WAV reanalysis are the mean of the differences between measured and computed values (bias) and the corresponding root mean square differences (RMSD). The results are also summarised in [DA2].

EANs are computed for:

- Significant wave height (SWH): Refers to the "spectral significant wave height (VHM0)".

The observations used for the comparison are:

- Significant wave heights (assimilated): recorded by the radar altimeters of the Jason-1 (JS1), Jason-2 (JS2), and Jason-3 (JS3) satellites and available on the public AVISO server[1], which additionally include wind speed measurements. These data have been merged into one dataset hereafter referred to as "JS merged".
- Significant wave heights (not assimilated): recorded by the radar altimeters of the Sentinel-3a/-3b (NRT), HaiYang-2b (NRT), Cryosat-2 (NRT+MY), CFOSat (NRT+MY), Saral/Altika (NRT+MY), and Envisat (MY) satellites. These data were obtained from the CMEMS NRT and MY wave altimeter datasets and were merged into a single dataset hereafter referred to as "All merged".
- Significant wave height: recorded by the available wave buoys for the considered period.
- Significant wave height and peak period: recorded by the Acoustic Doppler Current Profiler (ADCP) station Pasha Dere located at 28.03 °E, 43.08 °N.

The EANs computed for the v2024 version of the CMEMS Black Sea wave modelling system are based on the simulation of the system in hindcast mode for the 20-year period from January 2002 – December 2021 of the complete hindcast period (January 1950 to -18M). For the first 52 years of the hindcast period (between January 1950 and December 2001) there are no measurements available that could be used for a comparison. Given the lack of systematic in-situ measurements in the Black Sea, satellite measurements represent the main source of data to compare the wave model results. The final values for bias and RMSD are given in Table 1. Since bias refers to model mean minus the mean of the measurements, the EANs for the BLK-WAV system indicate a very slight underestimation of the measurements by the wave model for the comparisons with Jason-3 data. Please see Section VI for a comparison of model skill between the new model version (v2024) and the previous one (v2022).

[1] ftp-access.aviso.altimetry.fr (last accessed 13/11/2024) Note that an account (free) is required

Table 1: EANs for SWH for the BLKSEA_MULTIYEAR_WAV_007_006 reanalysis (all values in centimetres) for each of the three satellites named spanning different periods and for the case when all satellite measurements are combined (JS merged).

2002-2013 (Jason-1)		**2008-2019 (Jason-2)**		**2016-2021 (Jason-3)**		**2002-2021 (JS merged)**	
Bias	RMSD	Bias	RMSD	Bias	RMSD	Bias	RMSD
3	19	1	25	-1	25	2	23

EANs with respect to buoy data are available for a short period only and have not been included here. They will be included in a future update of this document.

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production centre details

Production centre name: Helmholtz-Zentrum Hereon (formerly HZG), Germany

Production system name: Black Sea Waves Reanalysis (E5R1)

Available variables (2D):

Datasets *cmems_mod_blk_wav_my_2.5km_PT1H-i*:

- VHMO: spectral significant wave height (Hm0);
- VTM10: spectral moments (-1,0) wave period (Tm-10);
- VTM02: spectral moments (0,2) wave period (Tm02);
- VTPK: wave period at spectral peak / peak period (Tp);
- VMDR: mean wave direction from (Mdir);
- VPED: wave principal direction at spectral peak;
- VSDX: Stokes drift U;
- VSDY: Stokes drift V;
- VHM0_WW: spectral significant wind wave height;
- VTM01_WW: spectral moments (0,1) wind wave period;
- VMDR_SW1: mean wind wave direction from;
- VHM0_SW1: spectral significant primary swell wave height;
- VTM01_SW1: spectral moments (0,1) primary swell wave period;
- VMDR_SW1: mean primary swell wave direction from;
- VHM0_SW2: spectral significant secondary swell wave height;
- VTM01_SW2: spectral moments (0,1) secondary swell wave period;
- VMDR_SW2: mean secondary swell wave direction from;
- VCMX: maximum crest trough wave height (Hc,max); and
- VMXL: height of the highest crest.

Dataset *cmems_mod_blk_wav-aflux_my_2.5km_P1H-i*:

- WSPD: equivalent 10-m wind speed derived from altimeter;
- WDIR: wind degrees from north;
- FRICV: friction velocity at sea water surface;
- CHNK: Charnock coefficient;
- Z0: surface roughness length;
- EFLX: energy flux into ocean;
- MFLXU: eastward wave momentum flux into sea water; and
- MFLXV: northward wave momentum flux into sea water.

Frequency of model output: Hourly instantaneous

Geographical coverage: 27.25°E → 42.00°E; 40.50°N → 47.00°N (the Azov Sea is excluded)

Horizontal resolution: 1/40° in zonal direction, 1/40° in meridional direction (ca. 2.5 km)

Vertical coverage: Surface only

Frequency of hindcast release: Reanalysis: once a year, Interim: monthly

Data assimilation: Yes, significant wave height and wind speed data from the satellites Jason-1, Jason-2, and Jason-3 (from AVISO).

Citation (DOI):

Dataset:

Staneva, J., Ricker, M., & Behrens, A. (2022). Black Sea Waves Reanalysis (CMEMS BS-Waves, EAS4 system) (Version 1) [Data set]. Copernicus Monitoring Environment Marine Service (CMEMS). https://doi.org/10.25423/CMCC/BLKSEA_MULTIYEAR_WAV_007_006_EAS4

The wave reanalysis for the Black Sea is produced by the Hereon Production Unit utilising the WAve Model WAM (described below).

The BLK-WAV system integration is composed of several steps:

1. Upstream data acquisition, pre-processing and control of: ECMWF ERA5 (atmospheric reanalysis) atmospheric forcing using one-hourly temporal and 0.25° spatial resolution and of the AVISO satellite data.
2. Upstream data acquisition and pre-processing of hourly sea surface height and surface currents from BLKSEA_MULTIYEAR_PHY_007_004.
3. Hindcast: WAM produces BLKSEA_MULTIYEAR_WAV_007_006.
4. Post processing: the model output is processed to obtain the products for the CMEMS catalogue.
5. Output delivery.

The BLKSEA_MULTIYEAR_WAV_007_006 production chain is represented in Figure 1. It shows as an example the MY production of the BLK-WAV datasets from Nov. 2023 (Y-1) to Nov. 2024 (Y). Dark green are the time series of the MY files contained in the *cmems_mod_blk_wav_my_2.5km_PT1H-i* and *cmems_mod_blk_wav-aflux_my_2.5km_P1H-i* datasets. The light green time series represent the Interim files contained in *cmems_mod_blk_wav_my_2.5km_PT1H-i* and the dashed fields periods updated outside of EIS.

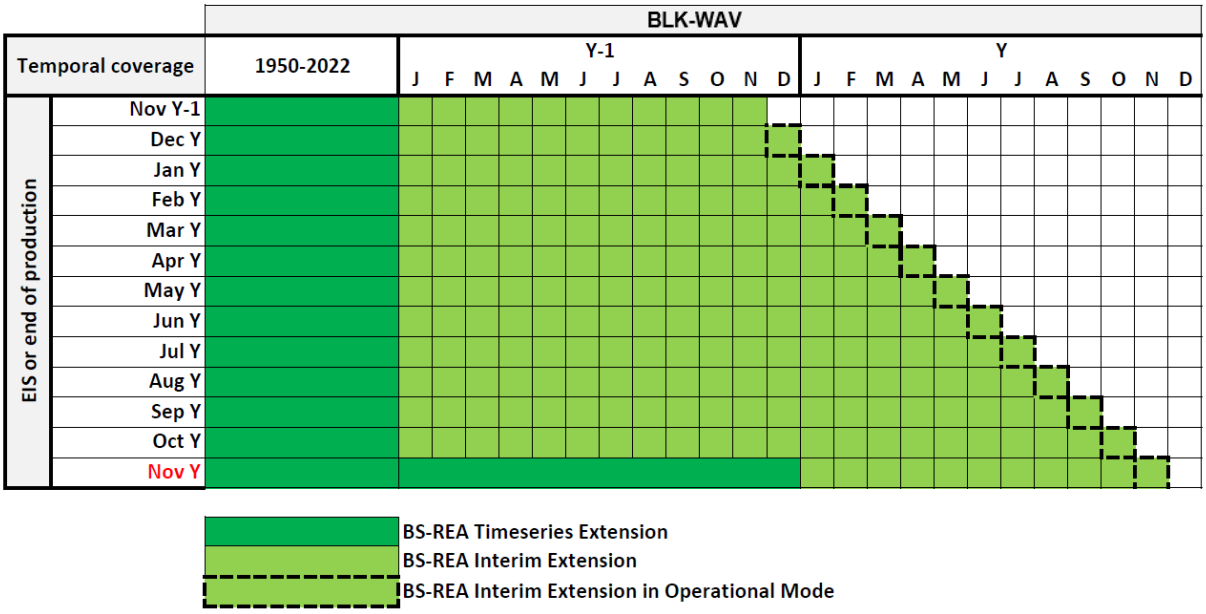

Figure 1: BLK-WAV production chain. EIS: Entry Into Service. Y: year.

II.2 System Description

This document details the quality of products from the Black Sea Wave Reanalysis system. These products are generated using the WAM Cycle 6 Black Sea model with a spatial resolution of about 2.5 km, which became operational within the Copernicus Marine Environment Monitoring Service (CMEMS) in April 2020 and was subsequently used to produce the BLKSEA_MULTIYEAR_WAV_007_006 reanalysis. The wave model describes ocean surface gravity wave (periods 1.5 to 24 seconds) characteristics as an extension to the existing physical and ecosystem model products provided by the Black Sea MFC. The following subsections describe the model component and its dependencies in terms of models providing the forcing.

Region, grid and bathymetry

The regional wave model for the semi-enclosed Black Sea runs in deep water mode on a model grid situated between 27.25°E to 42.00°E and 40.50°N to 47.00°N, with a spatial resolution of about 2.5 km, (1/40° in latitude and longitude direction). The required bathymetry for the model grid is based upon the General Bathymetric Chart of the Oceans (GEBCO 2019 Grid[2],) 30 *sec* data. The bathymetry is only a controlling mechanism on the wave field for depths below approximately 490 m, based on a minimum frequency in the model of 0.042 Hz (period of 24 s). The model area and the corresponding depth distribution are shown in Figure 2.

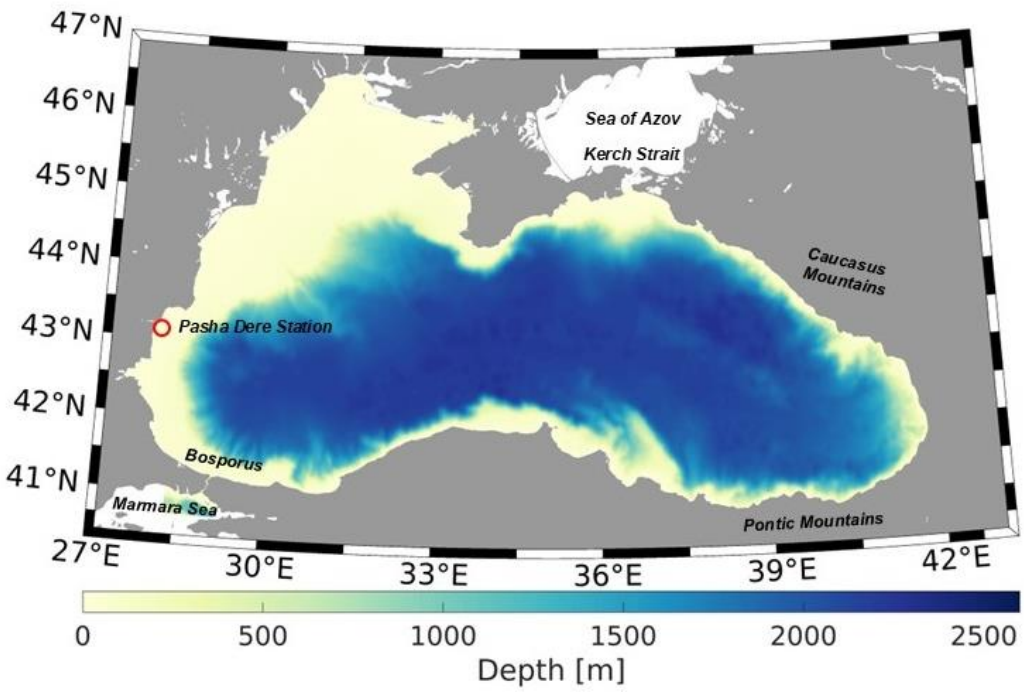

Figure 2: Black Sea MY wave model bathymetry including geographic locations and the position of the station Pasha Dere (red circle).

Spectral grid

WAM calculates the two-dimensional energy density spectrum at each of the 74,518 active model grid points in the frequency and directional space. The solution of the energy balance equation is provided for 24 directional bands each spanning 15°, starting at 7.5° and increasing clockwise with respect to true north, and 30 frequencies logarithmically spaced from 0.042 Hz to 0.663 Hz at intervals of $\Delta f/f = 0.1$. Therefore, the prognostic part of the wave model covers periods from approximately 23.8 to 1.5 seconds. In order to include the important contribution of higher frequency waves to wave growth/dissipation processes and for the output wave characteristics a parametric tail is fitted for frequencies above the spectral maximum (e.g., WAMDI Group, 1988).

[2] http://www.gebco.net (last accessed 01/11/2024)

Wave model and source term physics configuration

The system BLK-WAV is based on the state-of-the-art and well-established advanced third-generation spectral wave model WAM that runs successfully at many institutions worldwide. It is based on the spectral description of the wave conditions in frequency and directional space at each of the active model sea grid points of a certain model area. The energy balance equation, complemented with a suitable description of the relevant physical processes is used to follow the evolution of each wave spectral component. WAM computes the two-dimensional wave variance spectrum through the integration of the transport equation (1) in spherical coordinates, see also ECMWF (2020):

$$\frac{\partial F}{\partial t} + (\cos\Phi)^{-1}\frac{\partial}{\partial\Phi}(\dot{\Phi}\cos\Phi F) + \frac{\partial}{\partial\lambda}(\dot{\lambda}F) + \sigma\frac{\partial}{\partial\sigma}\left(\dot{\sigma}\frac{F}{\sigma}\right) + \frac{\partial}{\partial\theta}(\dot{\theta}F) = S \qquad \text{Eq.(1)}$$

$$\dot{\Phi} = (c_g\cos\theta + u_{North})/R \qquad \text{Eq.(2)}$$

$$\dot{\lambda} = (c_g\sin\theta + u_{East})/(R\cos\Phi) \qquad \text{Eq.(3)}$$

$$\dot{\theta} = c_g\sin\theta\tan\Phi/\mathrm{R} + \dot{\theta}_D + \dot{\theta}_C \qquad \text{Eq.(4)}$$

$$\dot{\sigma} = \dot{\sigma}_C \qquad \text{Eq.(5)}$$

with,

$\mathrm{F}(\lambda,\Phi,\sigma,\theta,\mathrm{t})$	wave energy density spectrum
(λ,Φ)	longitude, latitude
(σ,θ)	intrinsic frequency, wave direction

The first term of Eq. (1) describes the local rate of change of energy density in time, the second and third ones the propagation in geographical space, the fourth one the shifting of the relative frequency due to variations in depths and currents and the last one on the left side of the equation the contribution of the depth- and current-induced refraction. The source functions on the right of the transport equation comprise the contributions of wind input (S_{in}), nonlinear interaction (S_{nl}), dissipation (S_{dis}), bottom friction (S_{bf}) and wave breaking (S_{br}):

$$\mathrm{S} = S_{in} + S_{nl} + S_{dis} + S_{bf} + S_{br} \qquad \text{Eq.(6)}$$

A detailed description is given by the WAMDI Group (1988), Komen et al. (1994), Günther et al. (1992), and Janssen (2008). The WAM Cycle 6 that is used for the Black Sea wave hindcast is an update of the former WAM Cycle 4. The basic physics and numerics are kept in the new release. The source function integration scheme made by Ardhuin et al. (2010) is incorporated. The wave model performance has been discussed in Stanev et al. (2014), Staneva at al. (2015; 2016a, b), Wahle et al. (2015), and in the recent Ocean State Reports #4 & #5 (von Schuckmann et al., 2020, 2021). Wave breaking has been taken into account. Time dependent depth and current fields are not used in this setup.

Data assimilation

In contrast to the previous MYP, the assimilation of measured satellite data has been taken into account. The required radar altimeter data for that purpose is available on the public server of AVISO and includes SWH and Level- 2 (L2) wind speed data. The measured data are assimilated into the wave model fields using a sequential optimal interpolation (OI) scheme. The advantage of sequential methods, in contrast to variational methods, is their relative simplicity and computational efficiency. The disadvantage is the fact that SWH is not a dynamical quantity in the wave model, but only an output parameter derived from the spectrum. Therefore, the procedure is split into two steps: (1) the SWH and wind speed data are used to construct analysed fields of SWH and wind speed by the OI, and (2) the reconstruction of the spectrum from the analysed fields. Since the wave model generates one-hourly output, there may be a time difference of up to 30 minutes between the satellite observation and the output for the nearest model grid point. The spatial discrepancy can amount to half the model mesh size, i.e., about 1.25 km. Quality control is included in the model so that unreasonable values are discarded automatically. Satellites cross the Black Sea once or twice per day in under two minutes, so that very few measured values are available for assimilation into the wave and wind fields. Taking that into account and since the waves are propagating fast and have therefore a very small memory at that regional scale, we expect no significant influence of the assimilation on the wave model results. For the new MYP all radar altimeter measurements of the Jason-satellites that were available have been assimilated for the time periods that are covered by the three different Jason-satellites. Jason-1 data is assimilated for the first time period between 15/01/2002 and 21/06/2013, Jason-2 data between 22/06/2013 and 30/06/2019 and Jason-3 data between 01/07/2019 and 31/12/2021. Overlapping periods (04/07/2008 - 21/06/2013 and 12/02/2016 - 30/06/2019) are used for validation with independent satellite data.

Forcing

The driving forces for the wave model are the U10 wind fields provided by the atmospheric reanalysis ERA5 of the ECMWF (European Centre for Medium-Range Weather Forecasts). The U10 wind fields are disseminated through a corresponding ftp-server at a horizontal spatial resolution of 0.25°. The native spatial resolution of the ERA5 wind fields is 0.28124° which is about 31 km. The big advantage of the ERA5 U10 wind fields as the driving force for the waves is the one-hourly availability in contrast to the 6-hourly ERA-Interim wind fields. Hourly sea surface height and surface currents are used as additional forcing for the period 1993 to -18M, which are based on the BLKSEA_MULTIYEAR_PHY_007_004 product.

Boundary values are not considered. The Bosphorus is taken into account by including a small part of the Marmara Sea. The Kerch Strait is not considered and the influence of both straits is expected to be very small.

Wave growth

The growth of waves under extratropical wind storms and tropical cyclones has been the topic of several studies in the past two decades (Powell et al., 2003; Donelan et al., 2004; Zweers et al., 2010; Chen et al., 2013). Wave growth is controlled by the aerodynamic roughness of the surface, i.e., the drag that is felt by the wind. There is increasing evidence from theoretical (Makin, 2005), laboratory (Donelan et al., 2004) and field studies (Powell et al., 2003; Holthuijsen et al., 2012; Donelan, 2018) that the roughness (and thus the drag) starts to level off or even drop (Powell et al., 2003) at very high wind speeds. Thus, how wave models parameterize wave growth under high winds becomes increasingly important as forecast systems, both coupled and uncoupled, move towards higher resolution (Li et al., 2021, Breivik et al, 2022).

To reduce possible underestimates of satellite radar altimeter measurements by the wave model, the parameterisation of the wave growth in the wind input source term has been adapted to the driving wind fields. Wave growth is controlled by the wind input term (S_{in}). The form used in WAM Cycle 6 is based on the formulation presented as Eq.(7) in Ardhuin et al. (2010):

$$S_{in} = \frac{\rho_a \beta_{max}}{\rho_w \kappa^2} e^Z Z^4 \left(\frac{u_*}{c}\right)^2 \times \max(\cos(\theta - \Phi), 0)^p \sigma F(k, \theta). \qquad \text{Eq.(7)}$$

Here, F(k, θ) [m^3 rad^{-1}] is the wave variance density in wavenumber (k)-direction (θ) space, Φ is the wind direction, β_{max} is a constant nondimensional growth parameter, and

$$Z = \log(kz_1) + \kappa/(\cos(\theta - \Phi)(u_*/c + z_\alpha)) \qquad \text{Eq.(8)}$$

is an effective wave age with c the phase speed and z_1 a dimensionless wave age tuning parameter that shifts the growth curve. The directional spread is controlled by the power, p, a tunable constant which is commonly (and here) set to 2. Higher powers give a more narrowly directed wind input.

It is important to note that Ardhuin et al. (2010) already introduced a cap on the surface roughness in the form

$$z_0 = \min\left(\alpha_0 u^2/g, z_{0,max},\right) \qquad \text{Eq.(9)}$$

where g is the acceleration due to gravity and $z_{0,max}$ the maximum of z_0. The growth rate, normalised by the angular frequency ω, derived from a parametrization by Janssen (1991) results from:

$$\frac{\gamma}{\omega} = \varepsilon \beta x^2 \qquad \text{Eq.(10)}$$

with ε the air-water density ratio, β the Miles parameter, and

$$x = \frac{u^*}{c} \max\left(\cos(\theta - \Phi), 0\right). \qquad \text{Eq.(11)}$$

The Miles parameter β depends again on a constant called β_m with a default value of 1.2 (after Janssen, 1991). Here the β_{max} parameter has been tuned to $\beta_{max} = 1.8$ for the Black Sea to enable stronger wave growth.

In order to reduce possible underestimates of satellite radar altimeter measurements by the wave model, the parameterisation of the wave growth in the wind input source term has been adapted to the driving wind fields. The source term for the wind input is:

$$S_{in} = \gamma F \qquad \text{(wave growth × spectrum)} \qquad \text{Eq.(12)}$$

Partitioning method

Included in model outputs are characteristics describing individual wave components that make up a given sea-state. For example, a sea may consist simply of a single wind sea component for which all wave energy is affected by the forcing wind, or multiple swell components, which have been remotely generated by distant storms. In WAM these components are determined using a two-stage process. Individual components are derived from the two-dimensional wave spectrum. This process effectively treats the wave spectrum as a topographic map from which individual peaks in wave energy can be identified to define the separate wave components.

The second part of the procedure is based on the assumption that the wind sea component should be defined as only that part of the wave energy spectrum which is directly forced by wind (an assumption commonly used by operational wave forecasters who wish to directly compare the evolution of the wind sea component to changes in local wind conditions). Using this assumption, wave spectrum bins where

wave speed is slower than the (co-directed) wind speed are associated with the wind sea component. The assignment of special energy to the wind sea component overrides any previous assignment of wave energy to the topographic components made in the first step.

III VALIDATION FRAMEWORK

The skill and temporal variability of the 72-year reanalysis period included in BLKSEA_MULTIYEAR_WAV_007_006, spanning the period 08 January 1950 to 31 December 2021 is assessed by comparing the simulated significant wave heights with measurements provided by a set of satellites, namely Jason-1, Jason-2, and Jason-3 covering the reanalysis period from 2002 to 2021. For the comparison, we match significant wave heights from the satellites and BLKSEA_MULTIYEAR_WAV_007_006 in time and space. In addition, the CMEMS NRT dataset WAVE_GLO_PHY_SWH_L3_NRT_014_001 provided satellite data from Sentinel-3a/-3b, HaiYang-2b, Cryosat-2, CFOSat, and Saral/Altika. The CMEMS MY dataset WAVE_GLO_PHY_SWH_L3_MY_014_005 provided data from Cryosat-2, CFOSat, Saral/Altika, and Envisat. For the first 52 years between January 1950 and December 2001, there are no satellite measurements available to compare with.

For these comparisons, the satellite data had to be co-located in both space and time with the wave model data. Considering that the above satellites typically only require a maximum of two minutes to cross the Black Sea, their measurements were compared to the nearest model output time and grid point.

It should be noted that the satellite radar altimeter only provides measurements that could be used for the validation of the wave parameter significant wave height (SWH).

Although in-situ wave measurements from moored wave buoys are available from the CMEMS In-Situ Thematic Assemble Centre (CMEMS INS TAC) (INSITU_BLK_PHYBGCWAV_DISCRETE_MYNRT_013_034), the buoy locations are limited to coastal areas in the southwestern basin where water depths are mostly between 17 m and 20 m. In the model, these positions are located either right at land-sea boundary or only very few grid points away from the land boundary. As the wave parameters are very sensitive to the water depth, the model grid points used for the validation were slightly offset with regard to the buoy positions (by 1-2 grid points) in order to have more realistic water depths.

Periods with obvious buoy measurement errors were detected and excluded from the validations. Further information about the Black Sea buoy data quality is currently not available but a collaboration with INS TAC has been established to report measurement issues and allow the quality of the INS TAC data to be improved.

III.1 Quality control of the satellite data

Before making use of the satellite data, it need filtering to ensure physically plausible values for SWH. Even though the data provided by AVISO had been quality controlled, we found several cases that warranted further verification. Our quality control comprises manual and automatic filtering. At first, we screen the data to find obviously incorrect measurements, from which we derive criteria for automatic filtering. For instance, we remove all measurements, where the measured significant wave height equals 0. We also remove outliers that would possibly compromise our analysis. We define outliers as values that deviate by more than 2 standard deviations from the linear regression fit between measured and modelled data.

III.2 Statistical analysis

We present scatter plots of measured vs modelled significant wave heights for each satellite. We also show comparisons where all satellite data have been combined into a single dataset.

We avoid overplotting (as there are thousands of pairs of measured and modelled data) by estimating the bi-variate probability density by evaluating a 2D-Gaussian kernel on a square grid in the variable space.

Furthermore, the plots include summary statistics, such as the mean value and standard deviation, and statistics that describe the skill of WAM to simulate the significant wave heights.

The following skill scores are used: Pearson's product-moment correlation coefficient (Corr), root mean square differences (RMSD), bias, scatter index (SI), reduction of variance (RV), and the HH index (Hanna and Heinold, 1986). These scores are calculated as follows, where o and m stand for observed and modelled data, respectively. An overbar denotes a temporal average over the entire sample length n.

$$Correlation = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(o_i-\bar{o})(m_i-\bar{m})}{\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(o_i-\bar{o})^2}\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(m_i-\bar{m})^2}} \quad \text{Eq.(13)}$$

$$RMSD = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(m_i - o_i)^2} \quad \text{Eq.(14)}$$

$$bias = \frac{1}{n}\sum_{i=1}^{n}(m_i - o_i) \quad \text{Eq.(15)}$$

$$SI = \frac{\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(m_i-o_i-BIAS)^2}}{\bar{o}} \quad \text{Eq.(16)}$$

$$RV = 1 - \frac{\sum_{i=1}^{n}(m_i-o_i)^2}{\sum_{i=1}^{n}(o_i-\bar{o})^2} \quad \text{Eq.(17)}$$

$$HH = \sqrt{\frac{\sum_{i=1}^{N}(m_i-o_i)^2}{\sum_{i=1}^{N}(m_i o_i)}} \quad \text{Eq.(18)}$$

One of the underlying assumptions for the calculation of Corr is that variables follow a normal distribution, which is not the case for significant wave height. It might therefore be advisable to use a different measure for gauging the monotonic relation between modelled and observed significant wave heights, such as rank correlation. However, we use Pearson's correlation as it represents a quasi-standard for evaluating numerical models. Note that the skill metric, RV, is an adaptation of the Brier Skill Score and gauges the error variance (assuming a zero-mean error) against the variance of observations. RV is bound by $-\infty$ and 1, with 1 indicating perfect model skill and negative values indicating no skill at all. As the error variance, here expressed as the sum of squared errors, can be decomposed into components related to the standard deviations and the correlation coefficient, RV not only depends on the magnitude of errors, but also on the capability of WAM to simulate the temporal variability of the wave fields (Murphy and Epstein, 1989). In addition, the HH index is computed providing dedicated insights into the model performance when comparing to measurement data.

The scatter plots also show the least-squares linear fit (without including any intercept) between measurements and modelling results. Ideally, such a fit would be close to a 45° straight line from the origin, which has therefore been included as a reference.

Lastly, we also show pairs of quantiles of the measured and modelled significant wave heights. These quantiles are estimated from the empirical cumulative density function at specific percentiles. The highest quantile shown, corresponds to the sampled maximum value, which translates to the 100th percentile of the empirical distribution.

III.3 Time series analysis

The length of the period of the combined satellite measurements allows further assessment of the quality of the dataset through the computation of time series of statistical properties, which are derived on shorter time scales. We assume that monthly intervals already contain enough information to properly represent the Black Sea, although increasing time scales further improves the informational content. Short time scales are potentially affected by satellite data availability issues: typically a satellite takes less than two minutes to pass over the Black Sea, with fly-overs occurring once to twice daily. Consequently, very short-term statistics, e.g., using a daily scale, could not adequately represent the waves in the Black Sea waves. Therefore, we have chosen to use an annual scale for the validation.

We compute time series of specific quantiles, as well as mean, minimum, and maximum values of measured SWH for each year in the measurement period of the satellites. We apply this procedure to the corresponding simulated SWH from WAM for the comparison. The quantiles have been selected to represent the whole distribution of SWH with a focus on the upper quantiles. Similar to the scatter plots we provide the correlation, bias, and RMSD, but also include the relative error, which is obtained by

$$RErr = \frac{\frac{1}{n}\sum_{i=1}^{n}(m_i - o_i)}{\bar{o}} \qquad \text{Eq.(19)}$$

Note that this method heavily depends on the location and time of the satellite tracks and cycles. Likely, this approach does not catch all notable wave events. However, by using a relatively long interval to aggregate statistics, it can be assumed that the statistics are robust and, as a time series, can represent the wave climate of the Black Sea.

IV VALIDATION RESULTS

This section presents the validation of the BLK-WAV MY product. It is structured as follows. Section IV.1 shows overall metrics averaged in both space and time (the scatter plots) and only in space (the yearly time series) based on satellite altimetry data. Section IV.2 presents a first approach to map significant wave height error spatially, which is part of ongoing investigations. Section IV.3 comprises validations of several wave parameters such as wave hights, wave periods, and wave directions against in-situ buoy data. This is followed by a validation using a unique measurement of the Pasha Dere buoy covering a storm in Section IV.4. Sections IV.5 and IV.6 continue the satellite validation by showing spatially averaged waves heights and along-track storm validations, respectively. Section IV.7 demonstrates the importance of wind validations.

IV.1 Statistical analysis

The results of the comparison between satellite measurements and modelled significant wave heights is shown in Figure 3, which shows the scatter plots for the comparison between modelled and satellite-observed SWH. In general, all the computed and measured mean values of SWH are between 1.0 and 1.1 m. The bias is nearly zero for the merged satellite dataset, although the values for Jason-1 and Jason-2 show a very slight overestimation of the wave heights in WAM. The calculated biases of WAM for the different satellites confirm the good agreement between measurements and model results being in the range of -0.01 m (Jason-3) to 0.03 m (Jason-1). Noteworthy is that the full range of SWH shows almost no deviations from the 45°-line. RMSD, which is linked to bias, varies from 0.19 m (Jason-1) to 0.25 m (Jason-2 and -3). Another source of the deviations lies in the difference of the simulated and modelled variability, here given as the standard deviations. The differences range from about 0 cm (Jason-1) to 4 cm (Jason-3). Note that measurement errors and noise (that were not filtered out by our initial quality control) may also impact RMSD and the standard deviation of the measurements, thus potentially degrading the model skill. Additionally, the wind fields influence model variability and skill as they are used to force the wind-wave model WAM. The forcing fields reflect real atmospheric winds only to a certain extent regarding magnitude and variability as they come from the spatially coarse (0.25° x 0.25°) one-hourly ERA-5 reanalysis (see also Section IV.7).

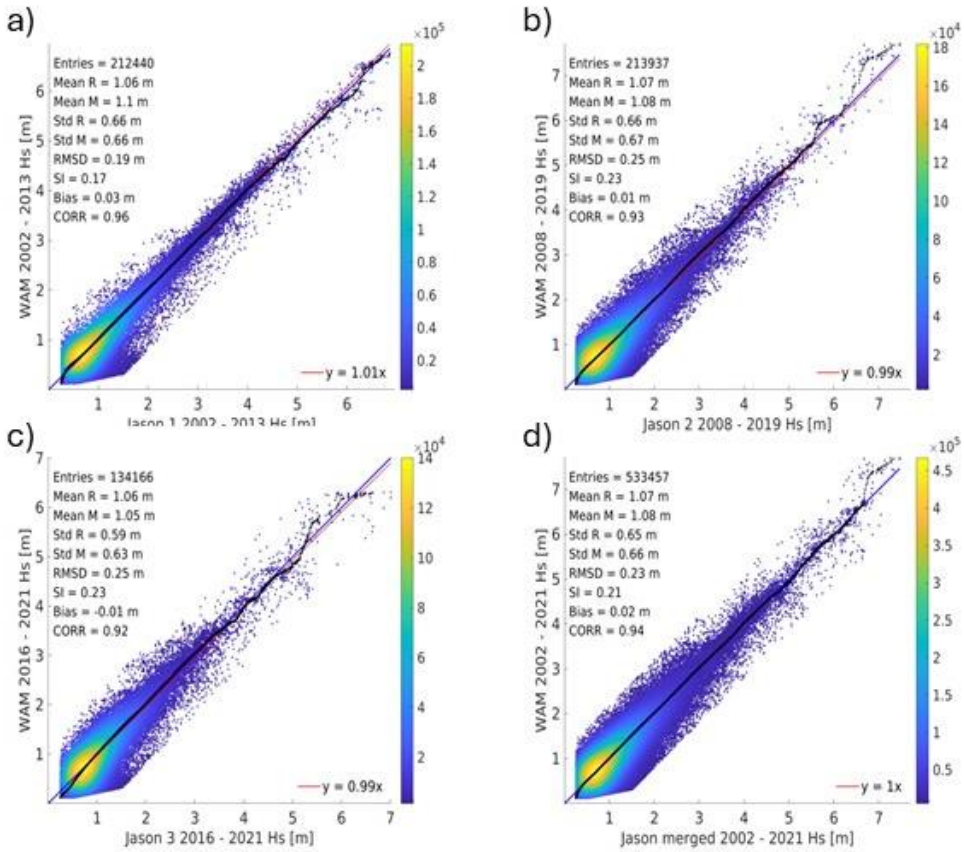

Figure 3: Scatter plots showing satellite measurements versus modelled significant wave heights for the periods a) 2002-2013 (Jason-1), b) 2008-2019 (Jason-2), c) 2016-2021 (Jason-3), and d) 2002-2021 (all Jason satellites merged). Also shown are the estimated bivariate probability density (colours), the least squares linear fit of modelled vs observed wave heights (red line), specific quantiles taken from the empirical cumulative density function (black line), and the 45° diagonal (blue line). Furthermore, summary statistics and skill scores are included. R: reference (satellite) data, M: model data, SI: scatter index.

However, despite these possible sources for uncertainty, the correlation, as a measure of how well modelled and observed SWH are positively linearly related, is in the range of 0.92 (Jason-3) to 0.96 (Jason-1), showing a strong linear relationship. Furthermore, the low scatter index (SI between 0.17 for Jason-1 to 0.26 for Jason-2 and 3) further attests to the high level of WAM model skill. Note that compared to skill assessments of short-term wave analyses and forecasts (see QUID for BLKSEA_ANALYSISFORECAST_WAV_007_003 for instance), the overall skill of the long reanalysis product presented here is slightly lower due to the longer period used for the comparison, which allows systematic biases, measurement errors, and noise to accumulate.

The good model skill in Figure 3 could also be related to the fact that the satellite data used for the comparison is assimilated into the model system. To obtain a more independent assessment, we compared the model data with non-assimilated satellite altimeter data from Sentinel-3a/-3b, Cryosat-2, CFOSat, Saral/Altika, and Envisat (Figure 4). Reassuringly, the model skill is nearly identical to the values obtained for the assimilated satellite data. RMSD is of the same magnitude (23 cm), SI is 2% worse, overall bias is now negative (-4 cm), and the correlation is 0.01 lower. As a result of the negative bias, the slope is 0.05 lower.

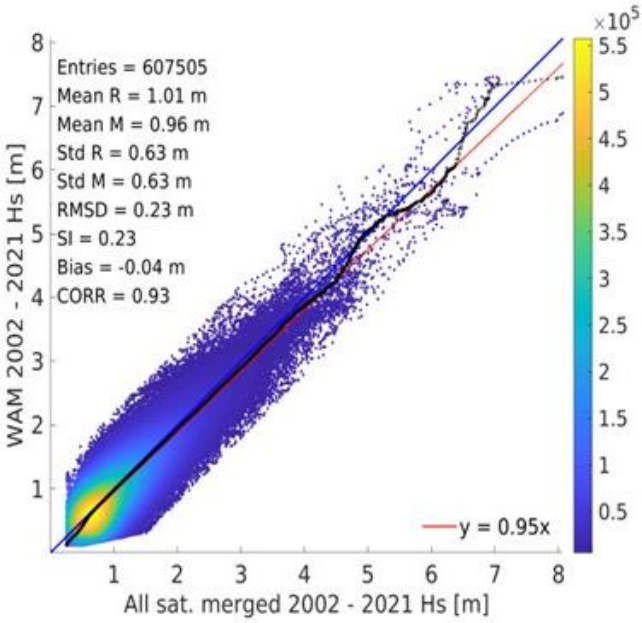

Figure 4: Scatter plot showing satellite versus modelled significant wave heights for the period 2002-2021 using the non-assimilated satellites Sentinel-3a/-3b (NRT), HaiYang-2b (NRT), Cryosat-2 (NRT+MY), CFOSat (NRT+MY), Saral/Altika (NRT+MY), and Envisat (MY) (data from individual satellites were merged). Also shown are the estimated bivariate probability density (colours), the least squares linear fit of modelled vs observed wave heights (red line), specific quantiles taken from the empirical cumulative density function (black line), and the 45° diagonal (blue line). Furthermore, summary statistics and skill scores are included. R: reference (satellite) data, M: model data, SI: scatter index.

Figure 5 shows time series of different statistical parameters for the complete validation period between 2002 and 2021 (using an annual scale). The metrics show a considerable interannual variability and also differ between satellites. In general, the following observations can be made: During the assimilation periods of the individual satellites, the corresponding metrics are relatively constant (RMSD: 18 cm, bias: 5 cm, correlation: 0.97, SI: 0.17, and RV: 0.92), whereas during those periods without data assimilation the quality of the statistics is slightly decreased. For example, RMSD increases to ~30 cm, bias decreases to -5 cm, correlation to 0.88, SI increases to 0.30, and RV decreases to 0.75. Nevertheless, even those metrics can still be considered as indicating good model performance.

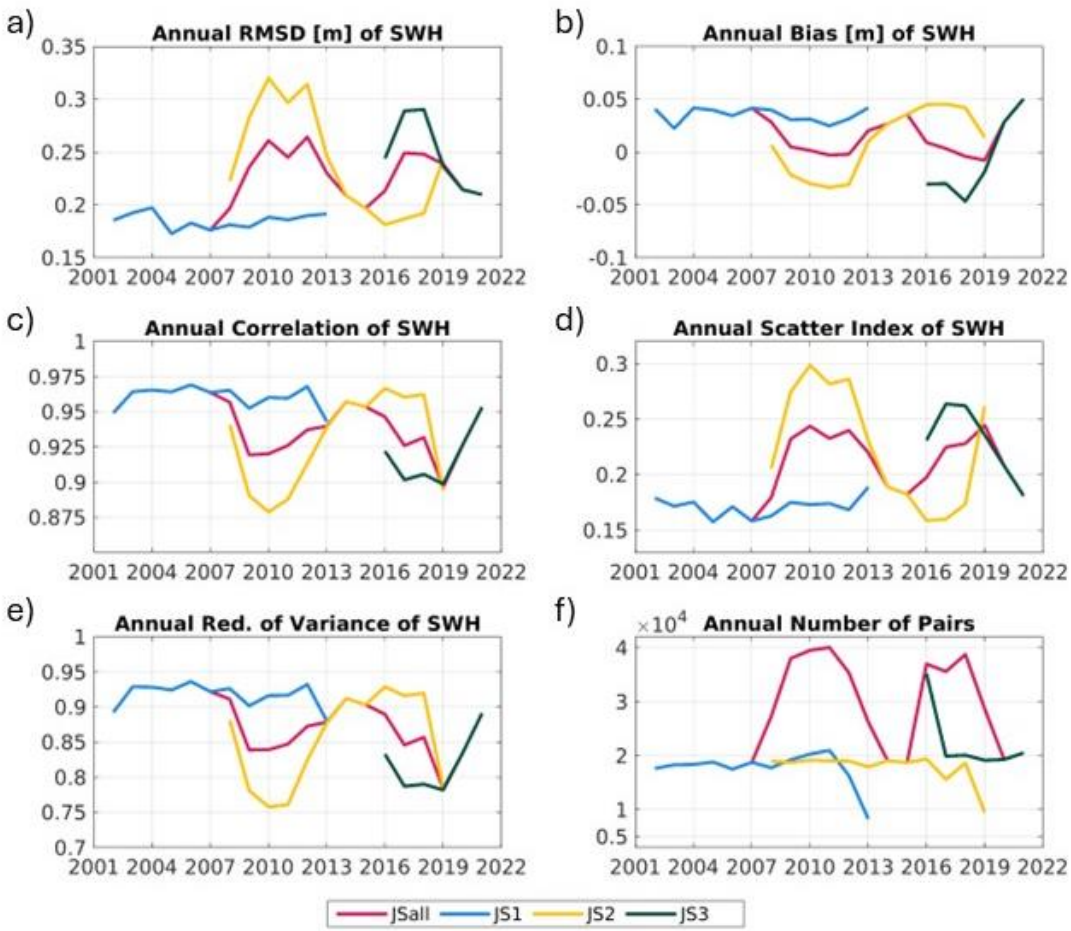

Figure 5: Annual significant wave height metrics for the Black Sea derived from individual and combined satellite measurements. "JSall" denotes the merged Jason data.

Overall, the skill scores depend on the number of collocated measurements to some extent. It is obvious that with a decreasing number of observations, parameters such as RMSD, bias, and scatter index increase, while RV and correlation decrease.

The time series of the 99th and 99.9th percentiles show a constant variability around 3.5 and 5 m respectively for both the model and satellite data (Figure 6). Also, for individual satellites the differences are equally low and large discrepancies only occur around 2013 when Jason-1 had reached its end-of-life.

The 99th and 99.9th percentiles of the SWH differences between the model and the satellites reveal a similar behaviour to the metrics in Figure 5; when the satellite data is assimilated, the percentiles decrease. When assimilated, only 1% of the differences are higher than 40 cm and 0.1% higher than 60 cm. If assimilated, these values increase to 70 and 100 cm, respectively.

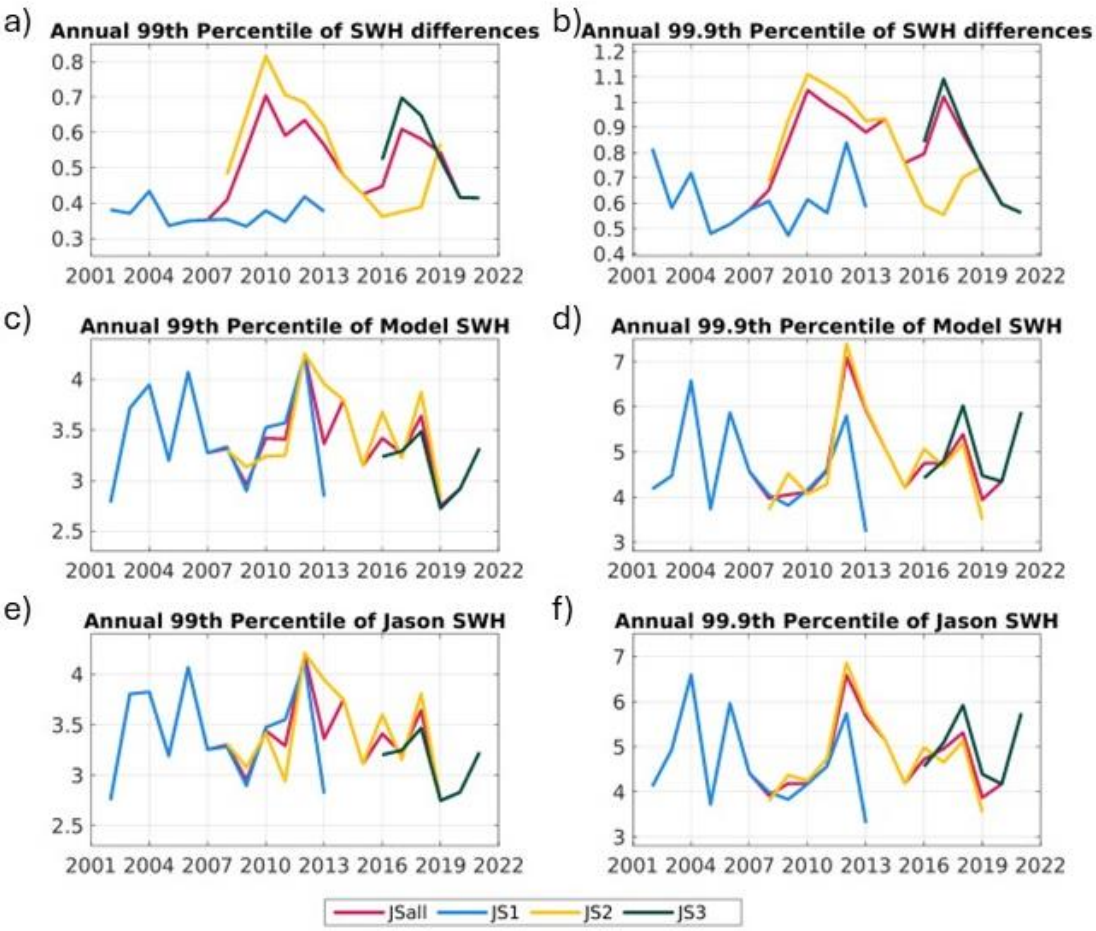

Figure 6: Annual percentiles for the differences in the significant wave height (a, b), the model wave height (c, d), and the wave height recorded by the radar altimeter of the satellites (e, f) (left: 99th percentiles, right: 99.9th percentiles). All units are given in m. "JSall" denotes the merged Jason data.

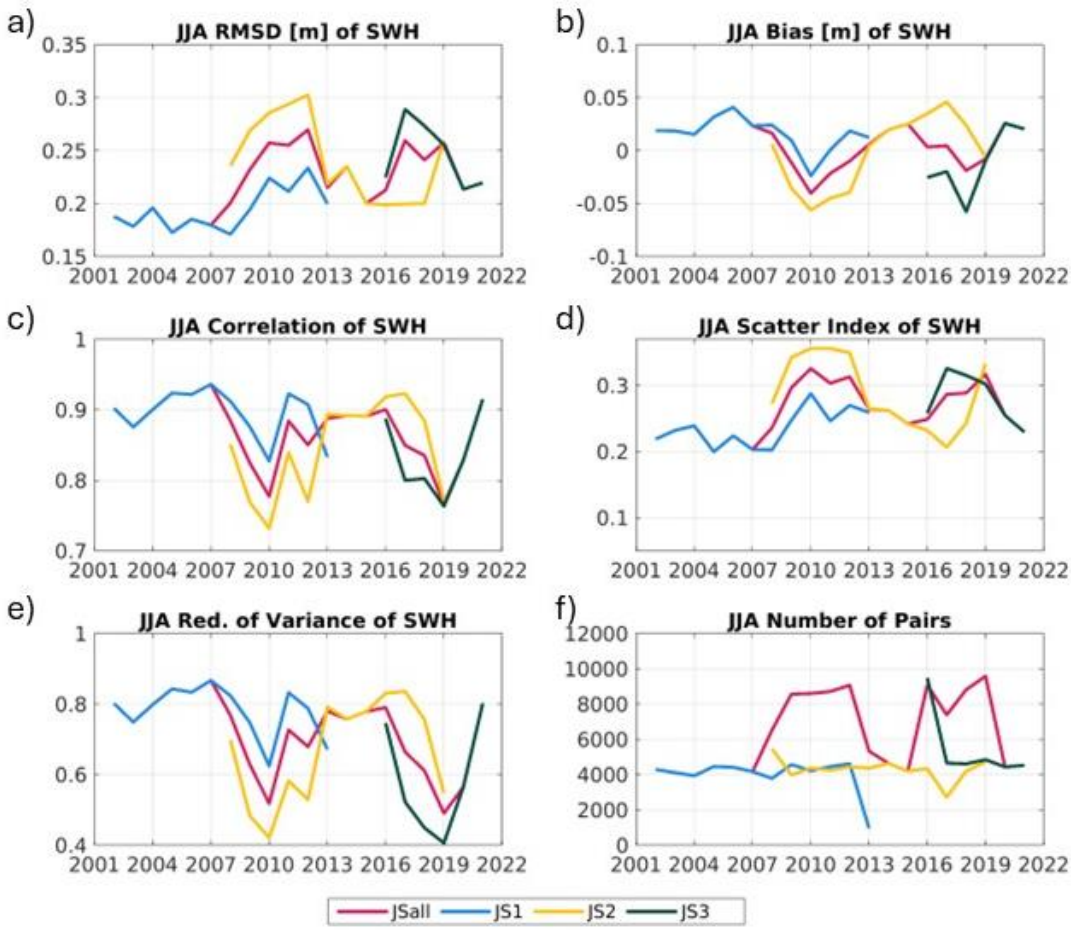

Figure 7: Annual values of summer (June through August - JJA) significant wave height metrics for the Black Sea derived from individual and combined satellite measurements. "JSall" denotes the merged Jason data.

The seasonal metrics shown in Figures 7 and 8 reflect the temporal evolution of the annual skill metrics. Some notable differences emerge. Firstly, the skill as measured by RMSD and bias seems better in summer (JJA, Figure 7) than in winter (DJF, Figure 8). RMSD (bias) ranges from about 0.17 to 0.30 m (-0.06 to 0.05 m) in JJA and from 0.16 to 0.35 m (-0.07 to 0.07 m) in DJF. Considering that winds are calmer during summer, wind-waves are consequently smaller in JJA, which improves skill metrics such as RMSD and bias. However, correlation and reduction of variance show opposite trends. In DJF, correlation (reduction of variance) ranges from about 0.74 to 0.94 (0.40 to 0.86) , while in JJA, the respective ranges are 0.86 to 0.98 and 0.70 to 0.97. Both metrics indicate that WAM model skill is lower in summer than in winter. Similarly, SI shows better values in winter (0.10 to 0.26) than in summer (0.20 to 0.34). This seasonality in the statistics is explained by seasonal differences in mean SWH, which tends to be lower in summer. In turn, lower mean values lead to worse correlation, SI, and RV if similar relative differences of model-satellite pairs are assumed, as they include temporal means (see Section III.2).

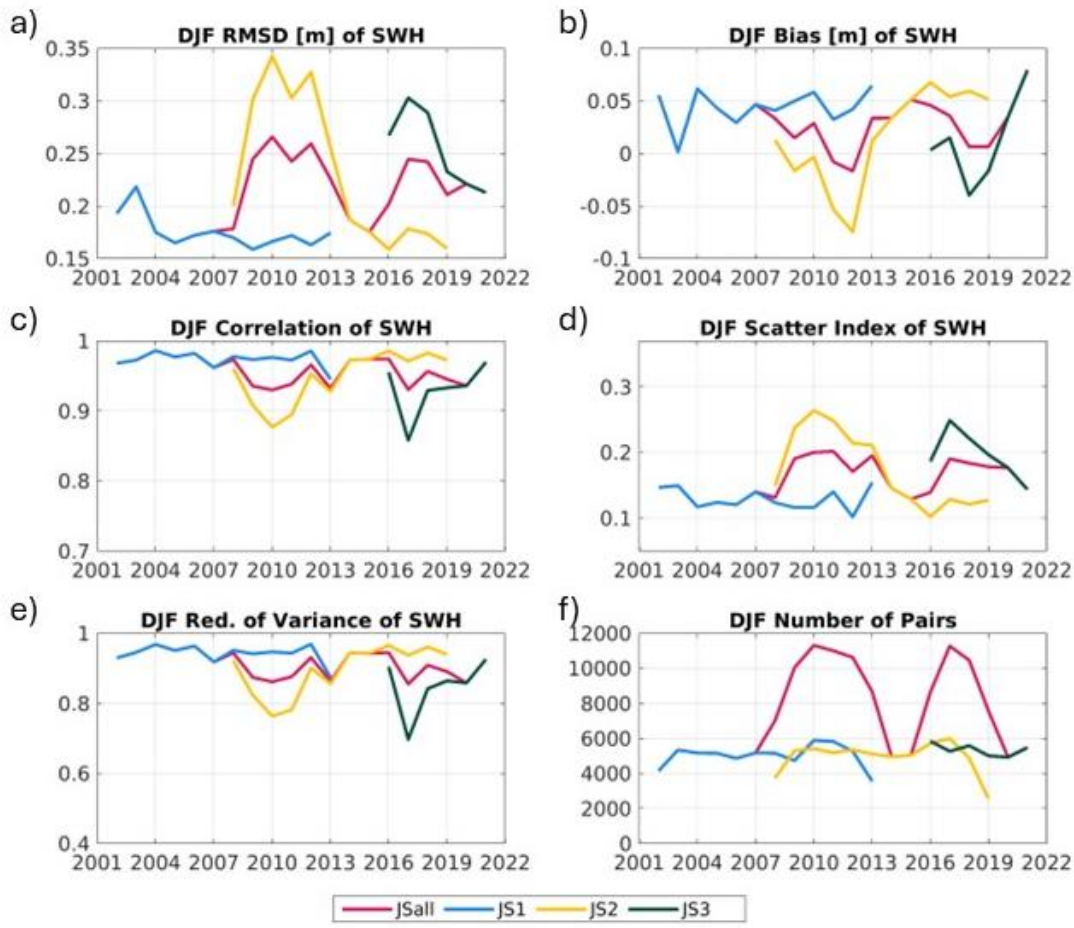

Figure 8: Annual values of winter (December through February - DJF) significant wave height metrics for the Black Sea derived from individual and combined satellite measurements. "JSall" denotes the merged Jason data.

IV.2 Error maps

The spatially varying metrics of significant wave height obtained from the same non-assimilated satellite data that were presented in Figure 4 are shown in Figure 9 below. The metrics are obtained by gridding the satellite data onto the model grid and performing the assessment grid-wise.

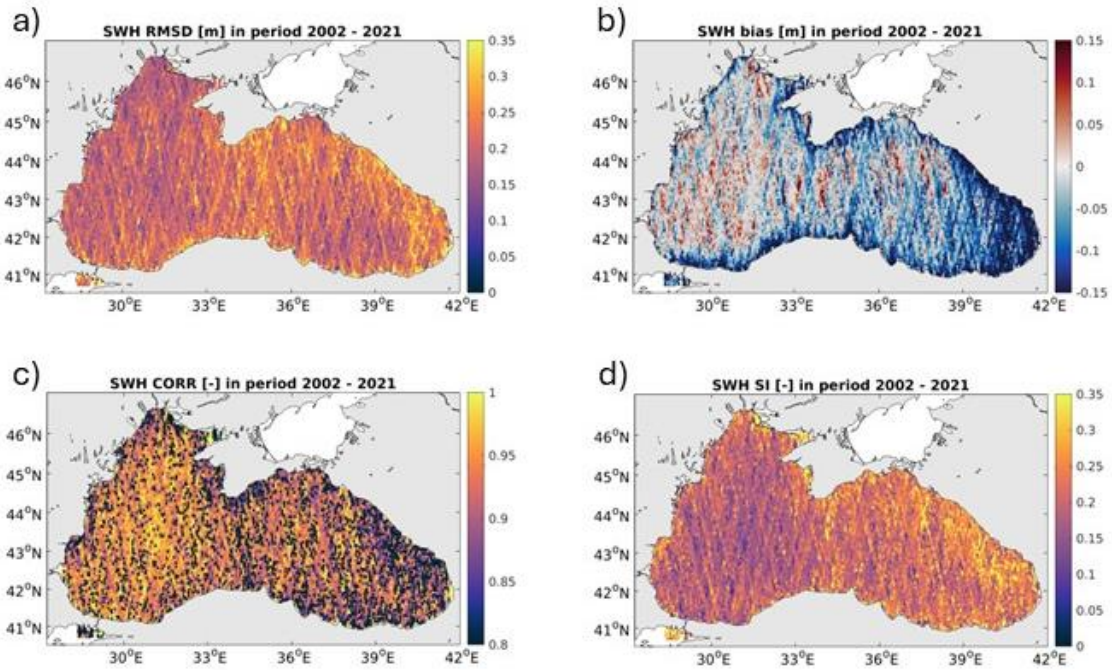

Figure 9: Spatially varying metrics of significant wave height obtained from the same non-assimilated satellite data presented in Figure 4. a) RMSD, b) bias, c) correlation, and d) scatter index. The maps are smoothed with a box filter.

Despite knowing the overall metrics and their seasonal changes, it is also important to determine these metrics for subregions of the domain to identify areas with unusually high or low model skills. A first attempt is presented in Figure 9 showing maps of the different model skill metrics with respect to non-assimilated satellite data. The results reveal that model skill is lower along the southern, eastern, and north-eastern coasts, e.g., bias is more negative. In contrast, model skill is better in the interior basins and along the western/north-western coasts. The individual metrics vary approximately as follows: RMSD ranges between 15 and 30 cm, bias between -15 and 5 cm, correlation between 0.80 and 0.97, and scatter index between 0.13 and 0.30.

IV.3 Buoy validations

The time period from May 2020 to December 2021 is (partly) covered by measurements of eleven wave buoys. Their locations and names are shown in Figure 10. The WAVEBXX and WD3044 buoys sit in a water depth of 17 m while the water depth of the SPOTXXXX buoys is 20 m. They provide the variables significant wave height, maximum wave height, wave mean direction, wave peak direction, wave T_{M02} period, and wave peak period (not consistent among all buoys). For a visual demonstration of the model performance, one buoy (WAVEB03) has been chosen and its results are presented in Figures 11-17 as time series. The available buoy time series cover the 6-month period from July 2021 until December 2021.

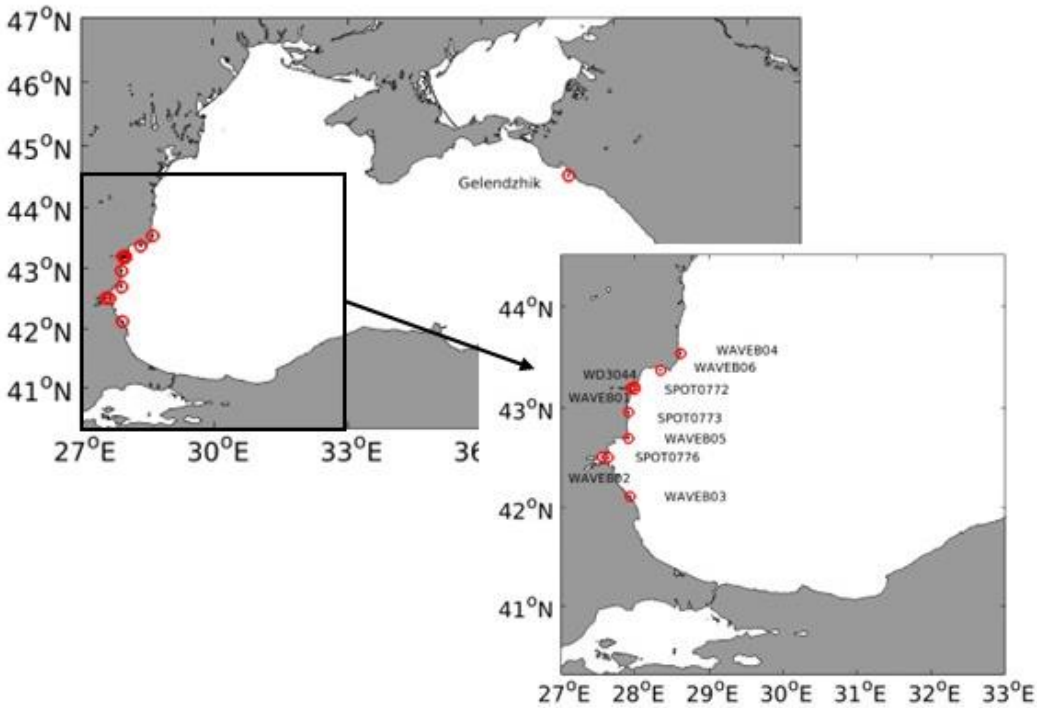

Figure 10: Locations in the south-western Black Sea of buoys with data during the NRT period.

For the case of significant wave height, the model performs well over the whole 6 months (Figure 11). The performance is good in both, SWH magnitude and the timing. The corresponding scatter plot (Figure 12) shows a relatively constant overestimation of the modelled SWH by 12 cm over the whole range of wave heights. It is likely that the proximity to the coast and uncertainties in the model bathymetry negatively influence the comparison. Nevertheless, the overall metrics are good with values of bias = 12 cm, RMSD = 20 cm, SI = 0.21, and Corr = 0.96.

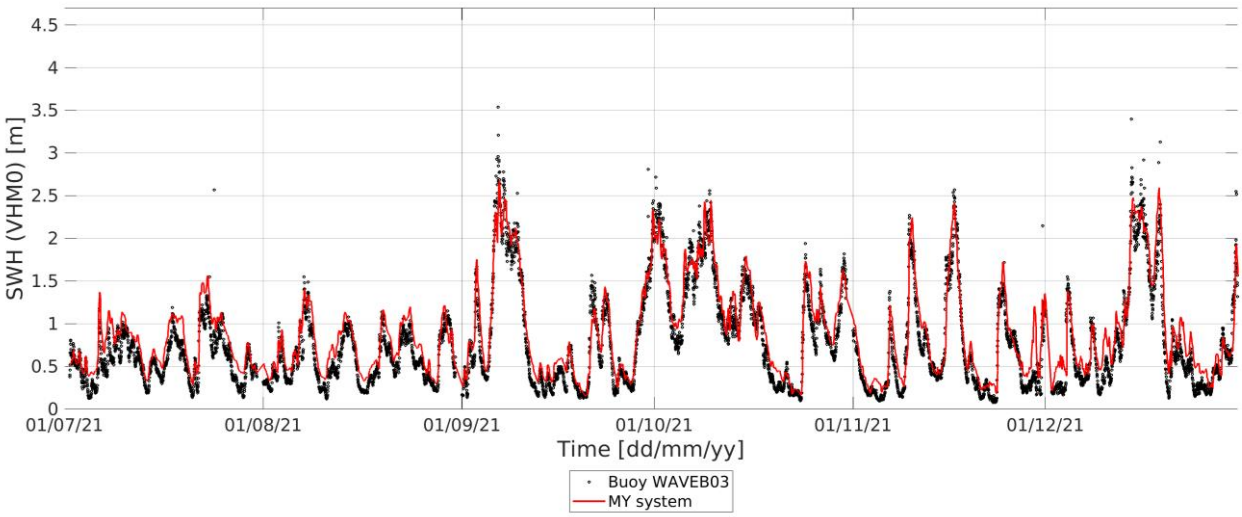

Figure 11: Comparing modelled significant wave height (VHM0) to in-situ data from the buoy WAVEB03 (all available data) for the period July to December 2021.

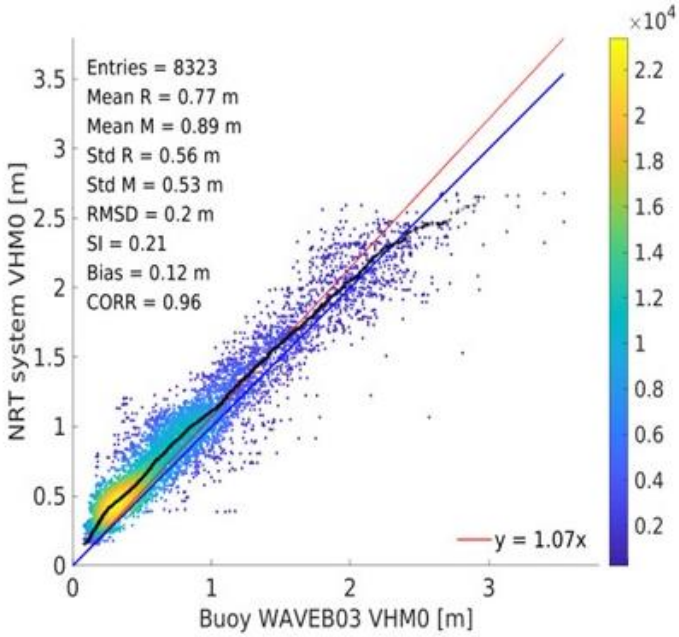

Figure 12: Scatter plot of significant wave height (VHM0) of the WAVEB03 buoy vs. model for the period) for the period July to December 2021 as shown in Figure 11. Also shown are the estimated bivariate probability density (colours), the least squares linear fit of modelled vs observed wave heights (red line), specific quantiles taken from the empirical cumulative density function (black line), and the 45° diagonal (blue line). Furthermore, summary statistics and skill scores are included. R: reference (satellite) data, M: model data, SI: scatter index.

The availability of in-situ observations made it possible to perform validations not only for SWH but also for other wave variables. Wave T_{M02} period (Figure 13) shows a similar behaviour as SWH. The timing is very good over the whole period but a slight overall underestimation is present. However, this

underestimation is mostly below 1 s (overall bias is -0.32 s), thus we can consider that the validations show good agreement with observations for the T_{M02} period. It should be noted that higher wave periods (> ~5 s) deviate more strongly from measurements (underestimated) than shorter periods.

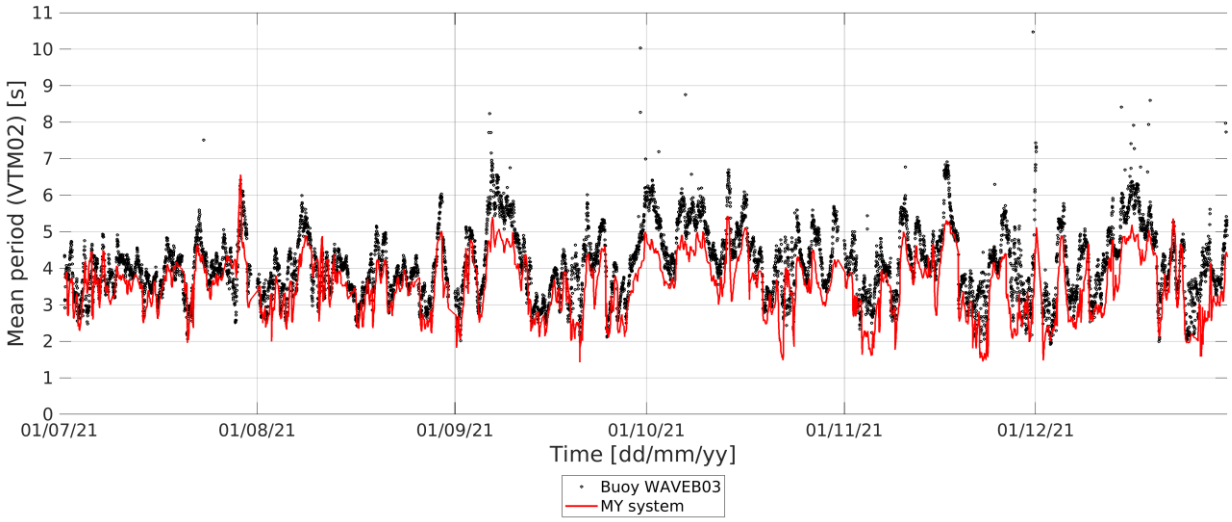

Figure 13: Comparing modelled wave T_{M02} period (VTM02) to data from the buoy WAVEB03 (all available data).

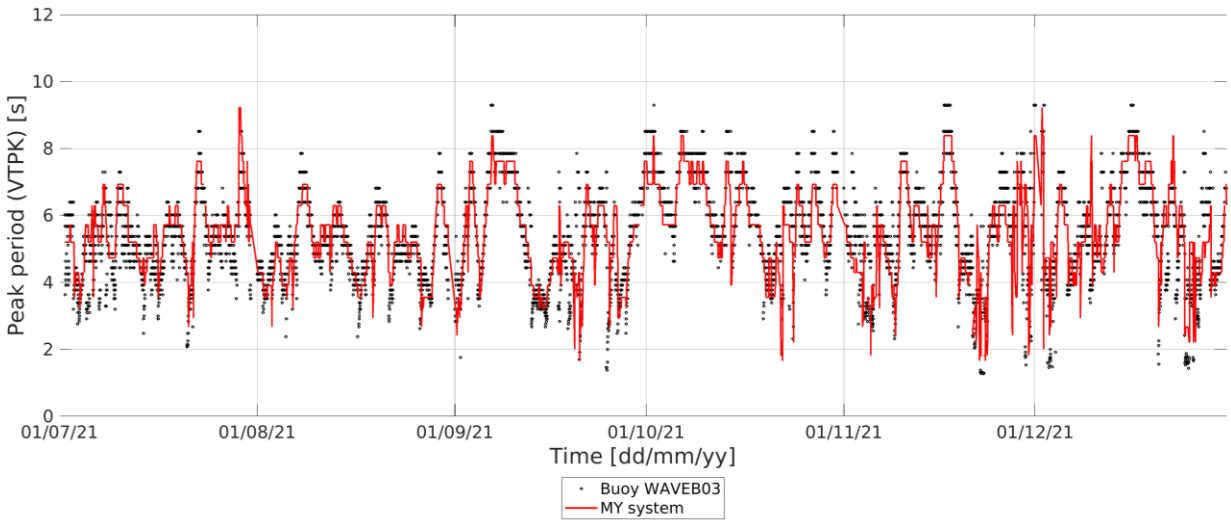

Figure 14: Comparing modelled wave peak period (VTPK) to data from the buoy WAVEB03 (all available data).

In contrast, the peak period is less underestimated (Figure 14) and the model performance for this variable is even better (overall bias is 0.02 s).

The simulated wave mean direction matches well with the in-situ data (Figure 15). Also, the short-term variations are well captured by the model. This good performance could be related to both the improved

spatial resolution and to the updated source terms. The performance of the wave peak direction (Figure 16) is comparable to the wave mean direction.

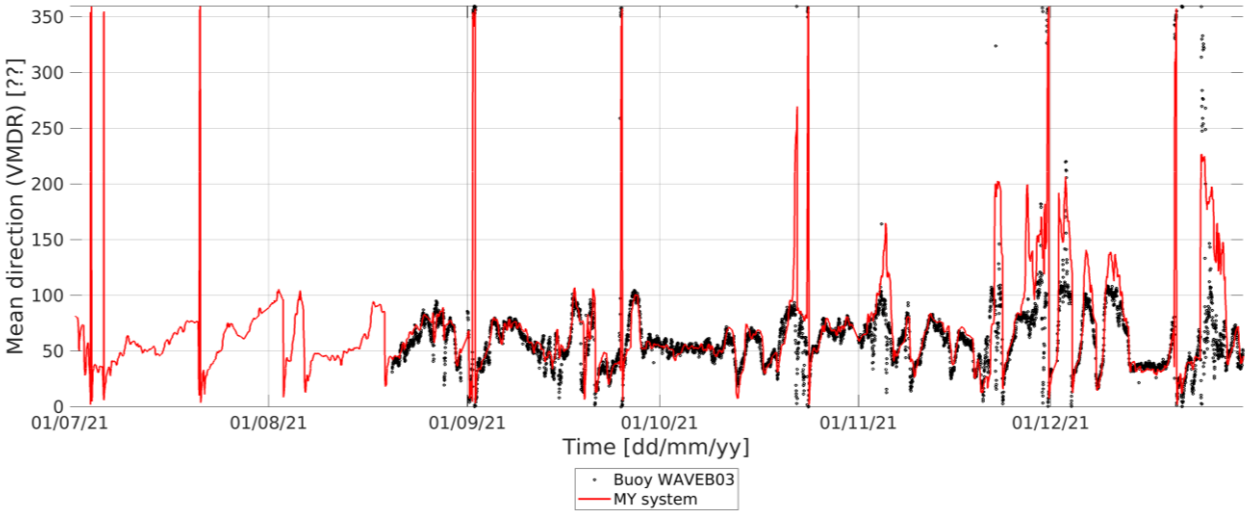

Figure 15: Comparing modelled wave mean direction (VMDR) to data from the buoy WAVEB03 (all available data). Vertical red lines appear due to the change from 359 to 0°.

Since the buoy WAVEB03 does not provide maximum wave height data, the results from buoy WAVEB01 (located further north in the Gulf of Varna, see Figure 10) we used instead for the comparison (Figure 17).

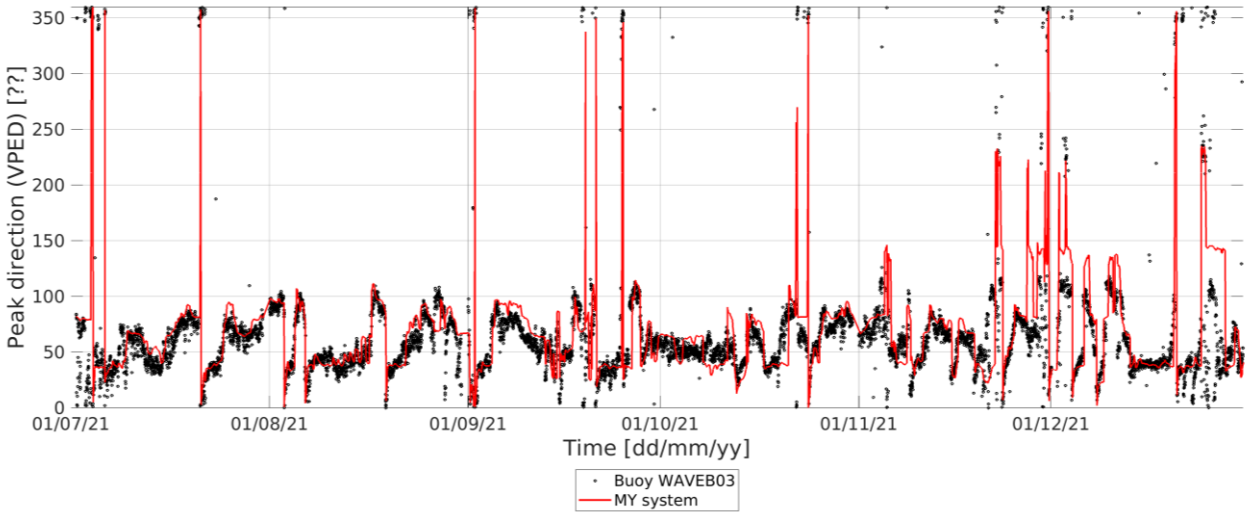

Figure 16: Comparing modelled wave peak direction (VPED) to data from the buoy WAVEB03 (all available data). Vertical red lines appear due to the change from 359 to 0°.

Compared to the significant wave height, the timing of Hmax is slightly worse. In addition, the maximum wave height simulation is also worse in terms of amplitudes, which could be due to the different type/location of the buoy. However, the overall model performance can still be considered as good.

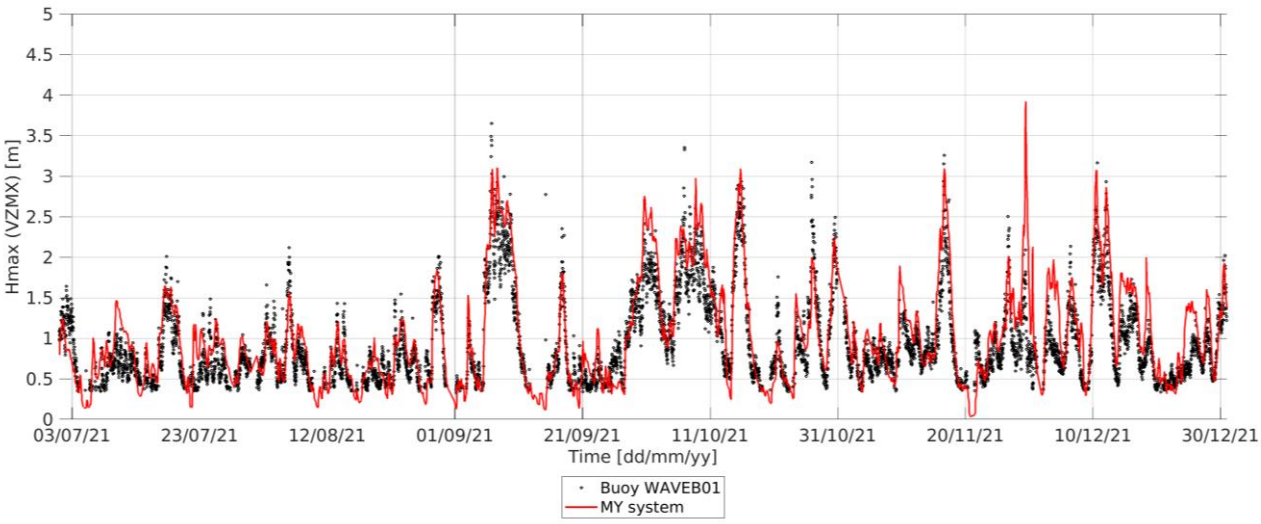

Figure 17: Comparing modelled maximum wave height (VZMX) to data from the buoy WAVEB01 (all available data). Hmax is not available for WAVEB03; instead WAVEB01 is shown.

When taking into account all buoys and all available buoy data for the considered period, the bias and RMSD of significant wave height are 10 cm and 20 cm, respectively (Table 2). As buoy data is not assimilated into the model, these values represent good metrics and also other variables like VTM02 and VTPK, show relatively slight deviations from measurements. Only VZMX shows a worse performance. Directional metrics will be introduced in the next issue of this QuID.

Although the model-buoy comparisons reveal good model skills, it is noteworthy that the buoy data contain periods with possibly reduced data quality. Especially, wave directions could be influenced by unprecise calibrations. As noted earlier, we established a collaboration with INS TAC to report and further investigate these measurement issues.

Table 2: Overall metrics for the MY BLK-WAV system using all buoys.

Period: May 2020 to December 2021		
Variable	**Bias**	**RMSD**
VHM0	10 cm	20 cm
VZMX	18 cm	41 cm
VTM02	-0.3 s	0.9 s
VTPK	0.0 s	1.3 s

IV.4 Comparison at location Pasha Dere – Storm representation

For the period between 04 February 2012 to 15 February 2012, ADCP data at station Pasha Dere (28.03°E, 43.08°N) in the western part of the Black Sea near the Bulgarian coast was available. This period covers a storm event occurring between 07 and 09 February 2012. The comparison with WAM for SWH and peak period is shown in Figure 18 with very good agreement. Although the skill scores for SWH are not perfect (bias: 24 cm, RMSD: 47 cm, correlation: 0.95, and SI: 0.31), the shape of the curve is often well captured. The corresponding values for the peak period, which depends on the resolution of the model frequency, are also very good with a bias of 0.2 s, RMSD of about 1 s, a correlation of 0.90, and SI of 0.14. Additionally, the storm peak was simulated quite accurately in terms of amplitude. However, the SWH decrease is slightly delayed, which is the reason for the worsened metrics. Nevertheless, this comparison illustrates nicely the capacity of the wave model to represent the arrival of a storm.

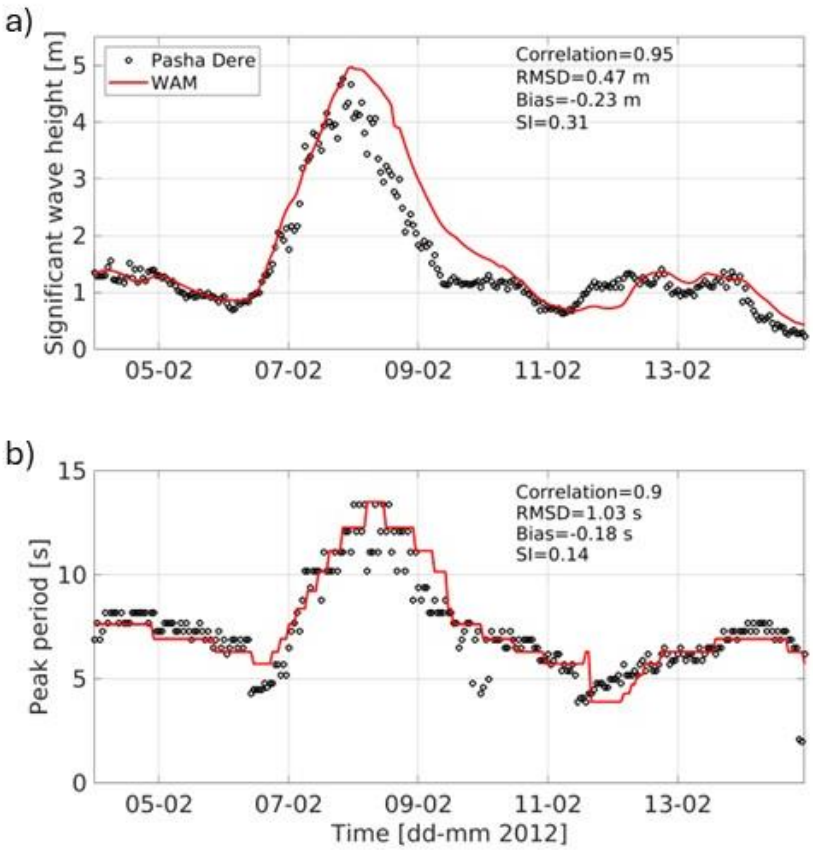

Figure 18: a) Significant wave height and b) peak period from WAM and from the ADCP station Pasha Dere in the period 04-15 February 2012.

IV.5 Time series analysis

Time series of wave statistics derived from satellites and WAM provide an opportunity to examine the temporal variability of specific parts of the SWH distribution and allow further model skill assessments. While the annual median modelled SWH varies between 0.82 and 1.02 m, the annual mean is slightly higher with values between 0.95 and 1.17 m (Figure 19a and b). Similar trends can be observed for satellite measurements (Figure 19c and d) median SWH ranging between 0.80 and 1.00 m annual means ranging between 0.92 and 1.13 m. An interesting feature here is the significant increase and subsequent decrease of the Jason-2 SWH near the end of its lifetime between the years 2016 and 2019. In contrast, Jason-3 which started in 2016, generates much smaller variations in SWH.

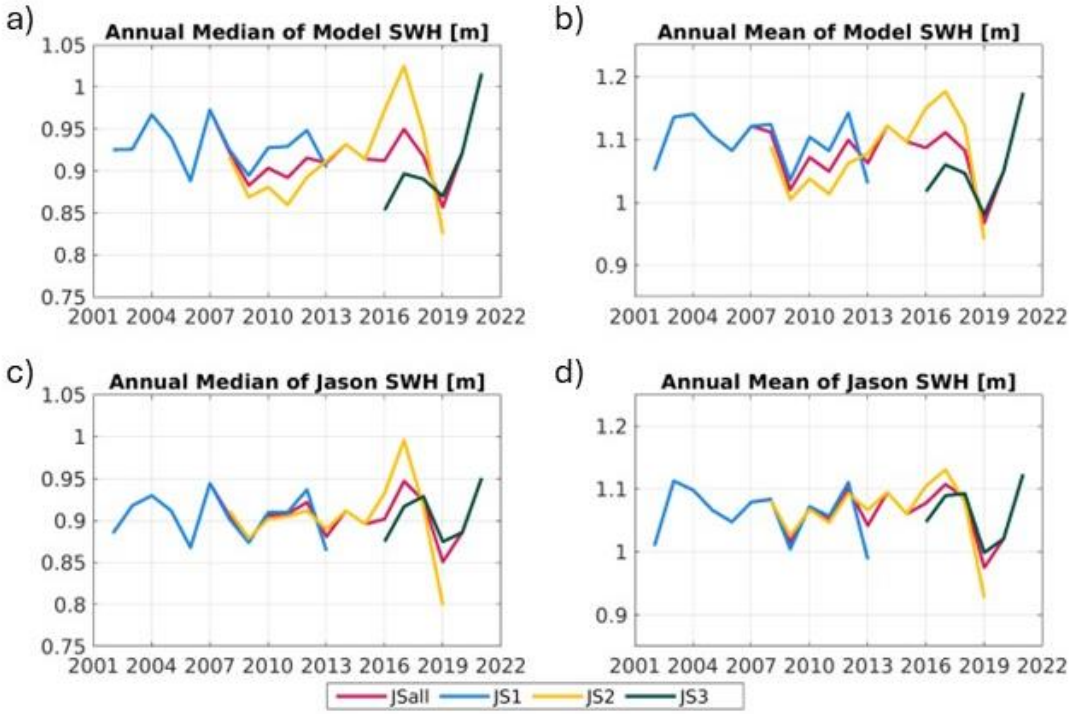

Figure 19: Time series of annual median (a, c) and annual mean (b, d) of the significant wave height along the satellite tracks shown for (a, b) model and (c, d) Jason satellite data.

If we examine annual values for SWH standard deviation and maxima, the model and satellites perform very similarly (Figure 20). Standard deviation values range between 0.53 and 0.75 m and annual maxima between 3.4 and 7.7 m.

As shown in Figure 21a to c, the annual mean SWH range from around 1.00 to 1.15 m without any obvious trend and with similar and sometimes slightly lower values from satellite measurements. The

lowest annual mean occurred in 2019, whereas 2021 was the year with the highest annual mean. The annual maximum SWH derived from satellite measurements and WAM show a correlation of about 0.97, which demonstrates the excellent WAM skill to simulate the temporal variability of maximum SWH.

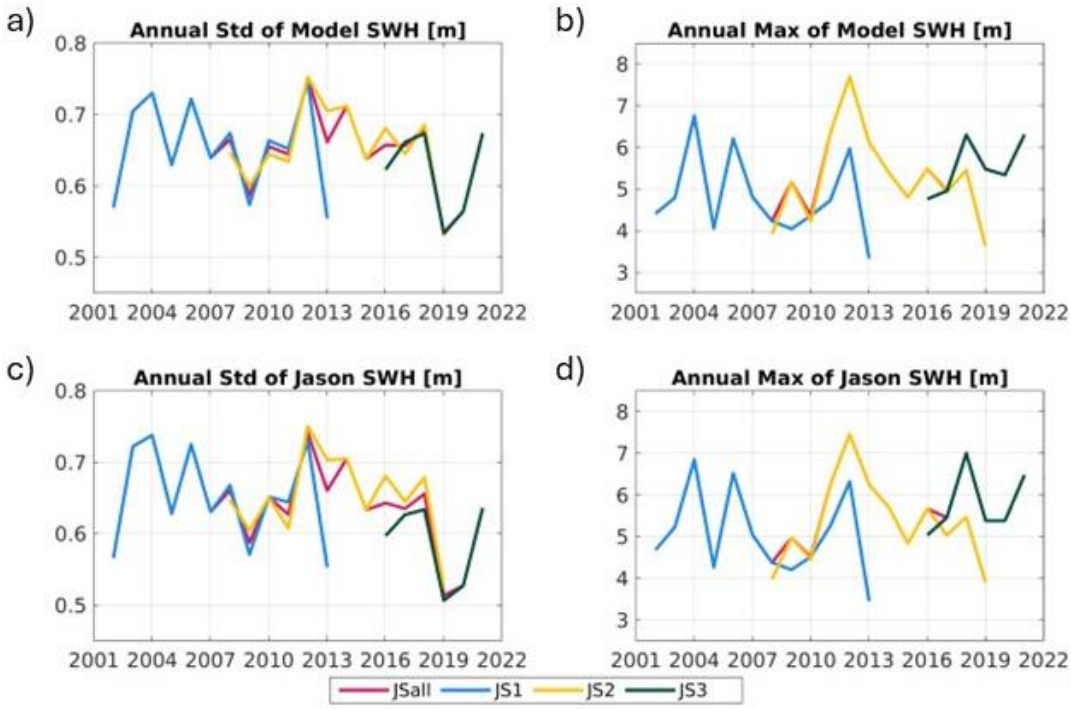

Figure 20: Time series of annual standard deviation (a, c) and annual maximum (b, d) of the significant wave height along the satellite tracks shown for (a, b) model and (c, d) Jason satellite data.

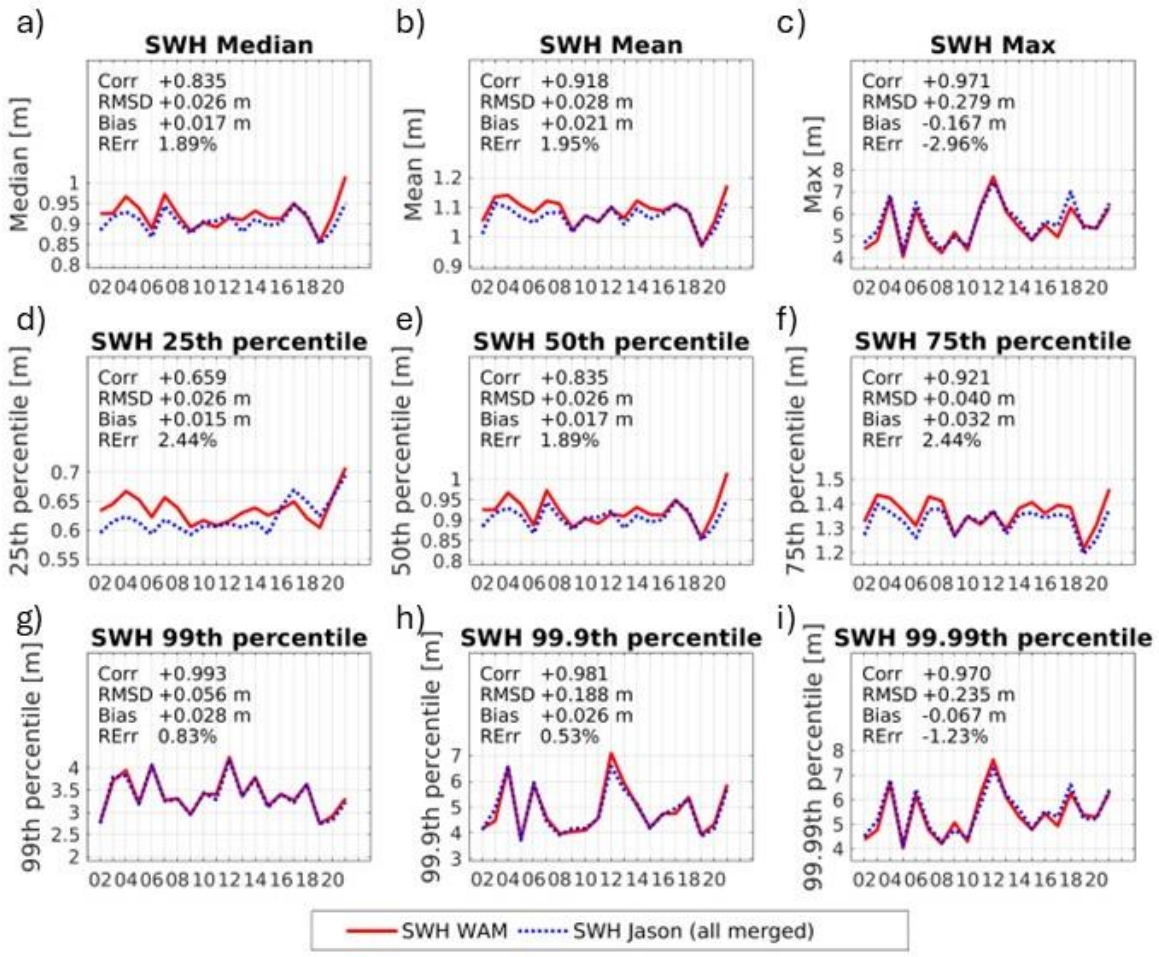

Figure 21: Time series of annual medians, means, and maxima of significant wave heights, as well as time series of specific annual quantiles of significant wave heights derived from all Jason satellite measurements and from WAM.

WAM slightly underestimates the maximum SWH resulting in a bias of -17 cm (corresponding to a relative error of -2.96% in Figure 21c). There are two years (2017 and 2018), for which maximum SWH is slightly underestimated. 2012 reveals the highest annual maximum SWH, which are also simulated quite well.

The annual quantiles of SWH from the 25th to the 99.99th percentiles (representing almost three-quarters of the distribution) show a temporal behaviour very similar to that of the annual means of SWH. The correlation between model and satellite data increases with higher percentiles reaching a maximum value of 0.99 for the 99th percentile. This demonstrates very good skill in capturing the variability of high SWH in the Black Sea. In contrast, the lower 25th, 50th, and 75th percentiles show lower correlations and are underestimated. The skill of WAM thus increases with higher percentiles, which possibly relates to parametrisations within WAM that slightly degrade the skill for smaller wave heights in favour of larger waves (von Schukmann et al., 2020).

While the annual 99th percentile shows a variability that is evenly distributed over the whole period with a very good agreement in time series behaviours, the 99.9th and 99.99th percentiles in WAM become dominated by a peak in 2012. These very high percentiles represent only a very small number of observations attributable to single events that do not markedly affect lower quantiles of the distribution.

Several factors might influence the assessment of SWH. First, the wind-wave model WAM, for which simulated wave heights depend on the wind forcing to a large extent, needs wind fields that include high wind speeds at the right time and space to model wave heights in agreement with satellite observations. Here, the used wind fields are taken from the ERA5 reanalysis, which provides data in a one-hourly resolution. It seems that this temporal resolution of the wind forcing is necessary to simulate the upper percentiles correctly.

Finally, the satellites measure wave heights along tracks at very fine temporal and spatial resolutions, i.e., much more detailed than the simulated wave fields of WAM. In our analysis, we match satellite measurements with their closest match in WAM introducing a sampling error that affects all parts of the wave distribution in our analysis. An even stronger effect could have the relatively coarse lateral resolution of the ERA5 wind fields.

IV.6 Along-track validations

Using a threshold of 3.5 m for SWH, Figures 22 - 24 show three example storm situations that occurred between 2002-2021. The first example describes the conditions on 8 February 2012 at 14:00 UTC showing the distribution of SWH and wave direction overlaid with the corresponding descending Jason-1 track crossing the south-western basin of the Black Sea and recording SWH up to 6 m (Figure 22). The along-track comparison Figure 22b illustrates the very good agreement between model and observations.

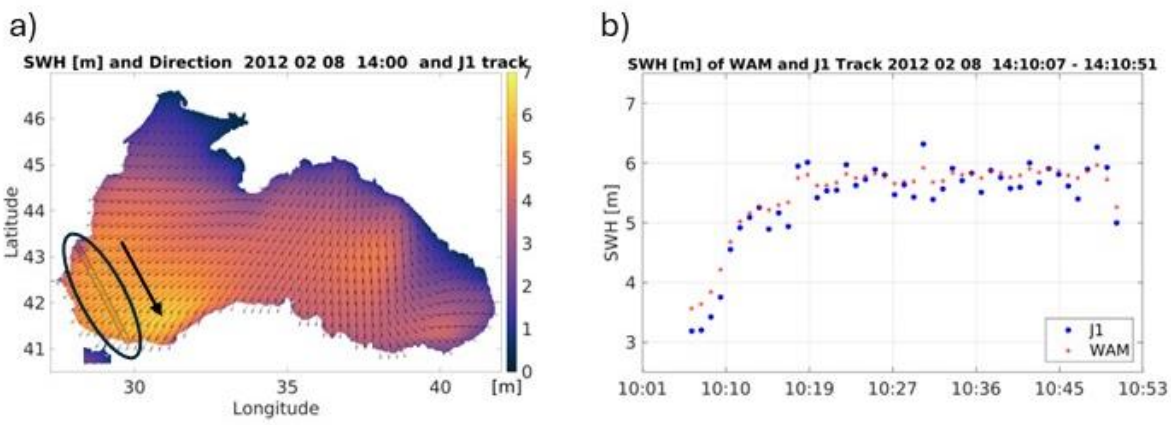

Figure 22: a) Significant wave height of WAM with overlaid Jason-1 measurements along the satellite track on 08/02/2012 14:00 UTC and b) time series of measured and computed significant wave height along the satellite track. Solid black ellipses: locations of the track. The x-axis represents time [mm:ss] for the day and hour given in the title. The arrow denotes the satellite flight direction

The second example shows an ascending Jason-2 satellite path that crosses a storm area in the eastern basin of the Black Sea (Figure 23). As in the previous example, the agreement between measurements and wave model is nearly perfect as demonstrated in the along-track time series (Figure 23b). The maximum SWH is around 5.5 m.

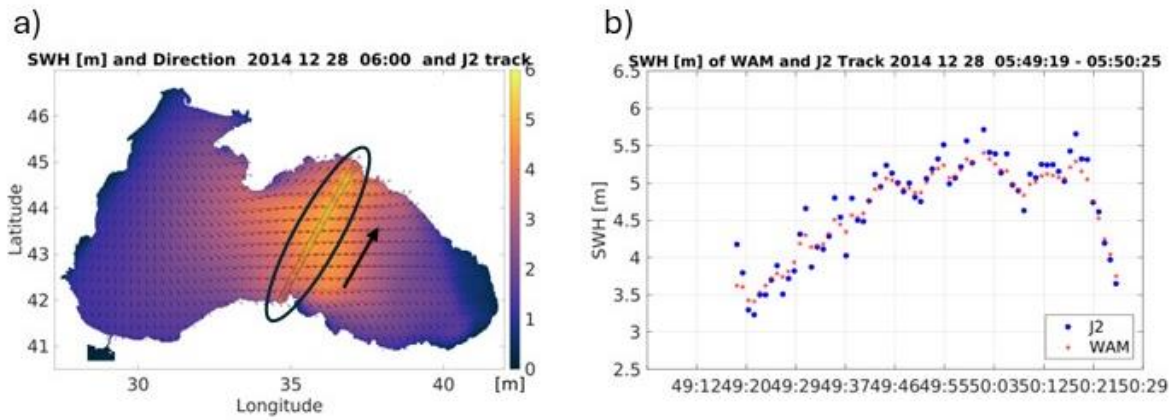

Figure 23: a) Significant wave height of WAM with overlaid Jason-2 measurements along the satellite track on 03 December 2016 06:00 UTC and b) time series of measured and computed significant wave height along the satellite track. Solid black ellipse: locations of the track. The x-axis represents time [mm:ss] for the day and hour given in the title. The arrow denotes the satellite flight direction

The third example shows a descending Jason-3 track passing a storm area in the eastern basin (Figure 24). As in the previous examples, the agreement between model and observations is again excellent. The maximum SWH is around 6 m.

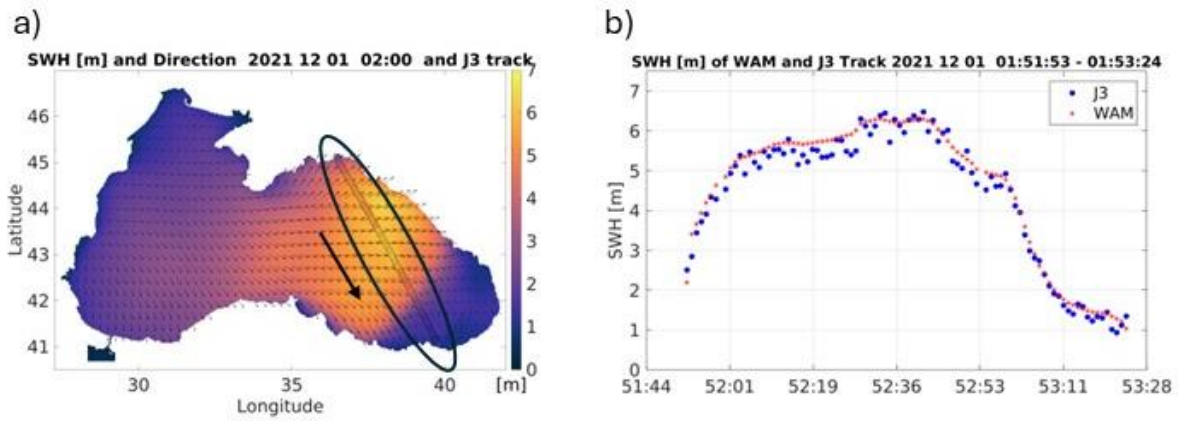

Figure 24: a) Significant wave height of WAM with overlaid Jason-3 measurements along the satellite track on 03 December 2016 06:00 UTC and b) time series of measured and computed significant wave height along the satellite track. Solid black ellipse: locations of the track. The x-axis represents time [mm:ss] of the day and hour given in the title. The arrow denotes the satellite flight direction.

IV.7 Wind validations

Wind forcing is the most important forcing type in the wave model. Thus, it is of utmost importance to determine its quality. As in-situ measurements of wind (e.g. from buoys) are lacking, the validation is restricted to satellite-obtained wind speeds. This quantity is also contained in the AVISO L2 satellite data. The assessment approach is the same as for SWH (see Figure 3). The comparisons are shown separately for each available satellite (Figure 25) as well as for all satellite data merged (Figure 25d). The wind forcing slightly overestimates moderate winds whereas low and high winds are represented quite well (see the qq-plot and the diagonal). Very high winds (>~12 m/s) tend to be too low in the forcing. The data of all satellites are quite close to the 45° diagonal and the qq-plots show relatively low deviations from the diagonal resulting in biases ranging from 0.06 to 0.26 m/s (Figure 25). RMSD ranges from 1.43 to 1.44 m/s with SI ranging from 0.23 to 0.24. CORR is also relatively constant with values of 0.85 to 0.86. For all satellites together, the respective values are: bias = 0.19 m/s, RMSD = 1.44 m/s, SI = 0.24, and CORR = 0.86. In summary, the quality of the wind forcing can be considered as good. However, the wind direction is not taken into account and is still an unverified error source.

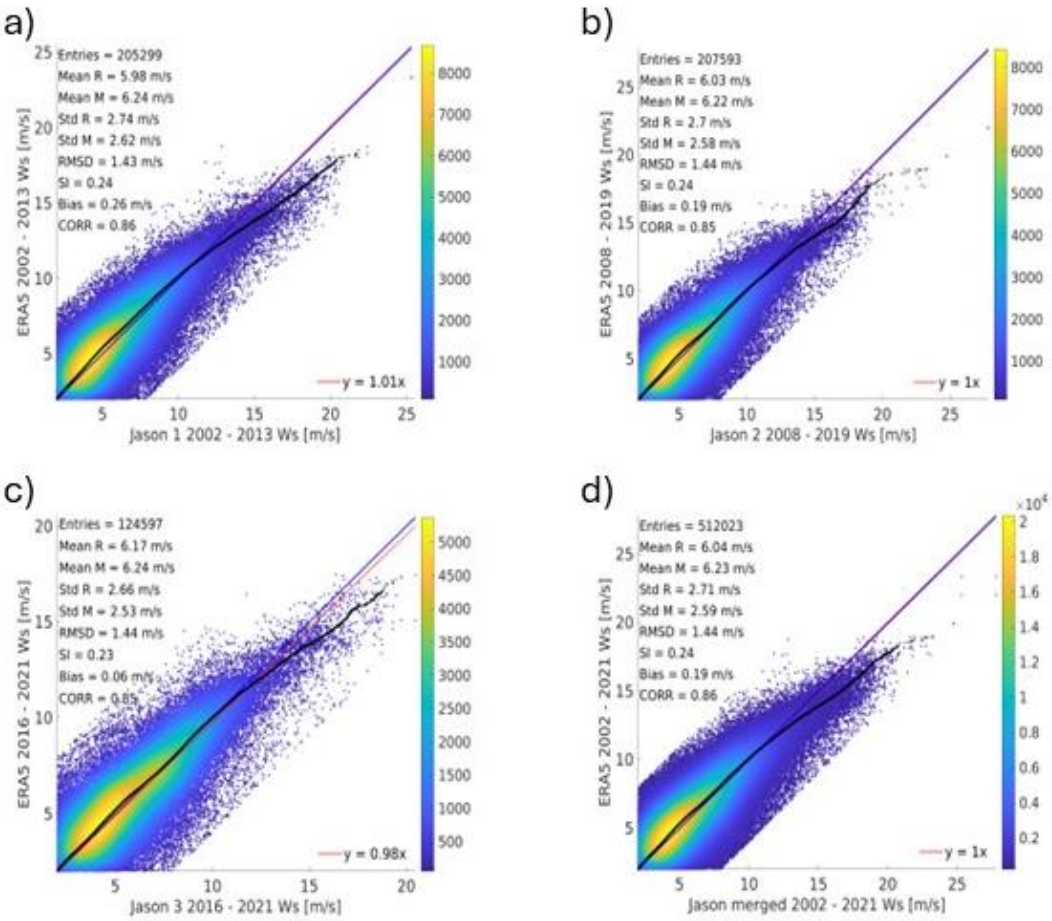

Figure 25: Scatter plots showing satellite measurements versus ERA5 wind speeds for the periods a) 2002-2013 (Jason-1), b) 2008-2019 (Jason-2), c) 2016-2021 (Jason-3), and d) 2002-2021 (all satellites merged). Also shown are the estimated bivariate probability density (colours), the linear slope-fit regression of modelled and observed wave heights (red line), specific quantiles taken from the empirical cumulative density function (black line), and the diagonal (blue line). Furthermore, summary statistics and skill scores are included. R: reference (satellite) data, M: model data.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. In includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

The old reanalysis BLKSEA_REANALYSIS_WAV_007_006 (v2022) has been replaced by the new model version v2024 of the reanalysis BLKSEA_MULTIYEAR_WAV_007_006 in November 2024. In addition, the new dataset *cmems_mod_blk_wav-aflux_my_2.5km_P1H-i* was added containing air-sea-flux variables.

Future time series extensions are expected to have no related changes in the quality of the product.

Interim files are operationally produced on a monthly basis since January 2021 covering the period from the end of the reanalysis files until M-1.

Date	Change/Event description	System version	Other
Sep 2019	Release of product	E1R1	
Dec 2020	New model version	E2R1	
Jan 2021	Addition of Interim product	E2R1	
Sep 2021	Addition of the 2020 time series extension	E3R1	
Nov 2022	MY dataset replaced (new model version WAM - Cycle 6, new model physics - ST4, updated bathymetry, higher horizontal resolution, 2 new maximum wave variables).	E4R1	
Nov 2023	Addition of time series extension to 1950 and addition of LATEMAR max. variables	E4R2	
June 2024	Removal of the cold start period (first week of 1950)	E4R2	
Nov 2024	Addition of sea surface height and sea surface currents from MY PHY as additional forcing and of a new air-sea-flux dataset	E5R1	
Nov 2025	Merging of the Interim and Reanalysis datasets	E5R1	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

Apart from the discussion of the quality of the new BLK-WAV system released in 2022 presented in the previous chapters, a very important question is how the new system performs in comparison to the previous wave routine (named "old"). Therefore, we present two of the previous plots (Figure 4) and compare them with the corresponding plots for the new model. Figure 26 and Figure 27 show the comparison between the old (v2022) and new (v2024) MYP based on the validation against the non-assimilated satellite and wave buoy data, respectively. In general, the statistical metrics of the new MYP system improved slightly compared to the old version. For the validation against the non-assimilated satellites, the RMSD reduced by approximately 3 mm, the Bias improved about 4 mm. Almost all other error metrics, such as SI, CORR, HH, and regression slope were also improved, but the differences are below practical significance of 1 cm/0.01. For the validation against the wave buoy data, the RMSD improved by about 5 mm, but the difference in Bias was increased by 0.5 mm. The differences of other metrics are also minor. All statistical metrics are presented in Table 3 and Table 4, corresponding to the validation against satellite and wave buoy data, respectively. Due to the insignificant improvements and the only partial replacement of the time series, the v2022 assessment of the previous sections are still valid as the v2024 results would be practically the same.

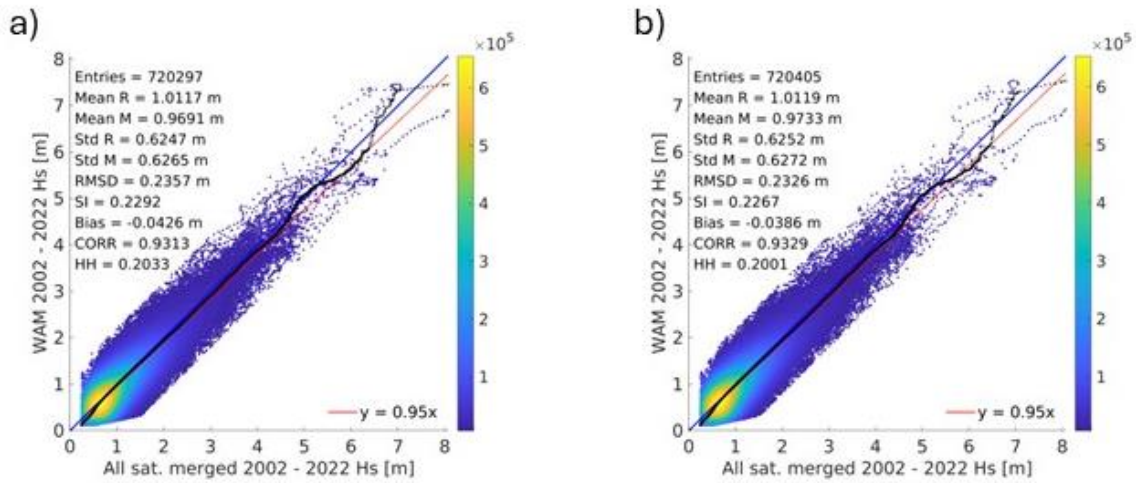

Figure 26: Scatter plot showing satellite versus modelled significant wave heights for the period 2002-2022 using the non-assimilated satellites Sentinel-3a/-3b (NRT), HaiYang-2b (NRT), Cryosat-2 (NRT+MY), CFOSat (NRT+MY), Saral/Altika (NRT+MY), and Envisat (MY) (data from individual satellites were merged); a) old MYP system (v2022), b) new MYP system (v2024).

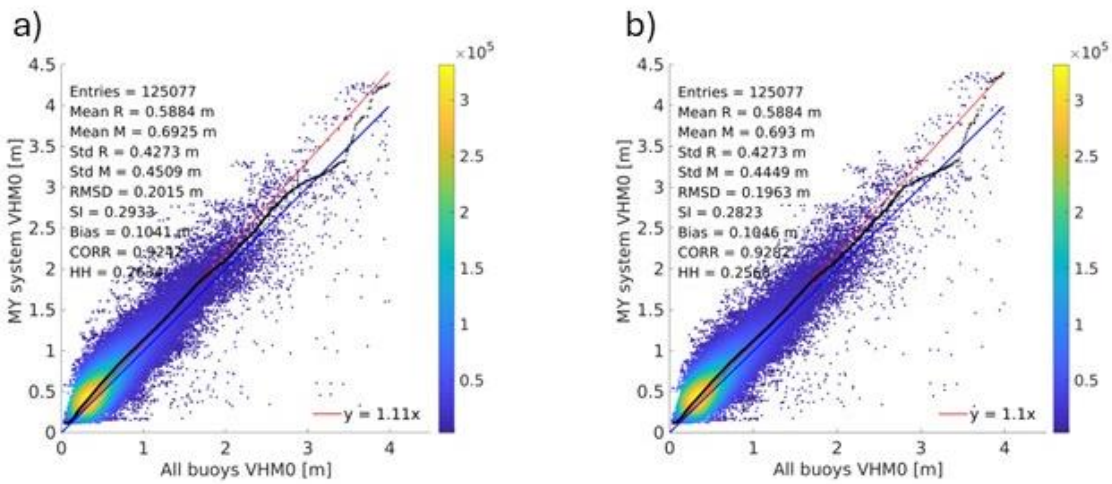

Figure 27: Scatter plot of significant wave height (VHM0) of all buoy data vs. model, a) old MYP system (v2022) and b) new MYP system (v2024).

Table 3: Statistical metrics of model validation against non-assimilated satellite data. Bold numbers represent an improvement of the new v2024 version, italic a deterioration.

	RMSD (m)	Bias (m)	SI	Corr	Slope	HH
Old (v2022)	0.2357	-0.0426	0.2292	0.9313	0.95	0.2033
New (v2024)	0.2326	-0.0386	0.2267	0.9329	0.95	0.2001
Difference	**0.0031**	**0.0040**	**0.0025**	**0.0016**	0.00	**0.0032**

Table 4: Statistical metrics of model validation against wave buoy data. Bold numbers represent an improvement of the new v2024 version, italic a deterioration.

	RMSD (m)	Bias (m)	SI	Corr	Slope	HH
Old (v2022)	0.2015	0.1041	0.2933	0.9242	1.11	0.2634
New (v2024)	0.1963	0.1046	0.2823	0.9282	1.10	0.2568
Difference	**0.0052**	*-0.0005*	**0.0110**	**0.0040**	**0.01**	**0.0066**

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VIII ANNEXES

In the following sections, an assessment of the Black Sea mean sea state is presented as they provide common metrics and variables often required by users. Section VIII.1 provides time-averaged spatial maps. Section VIII.2 comprises histograms of the significant wave height, Section VIII.3 presents the differences of the eastern and western Black Sea basins, and Section VIII.4 illustrates the content of the climatology available as a dataset of this product.

VIII.1 Mean Sea State

When looking at long term averages, the highest mean SWH are found in the southwest with values of up to 1.1 m on average (Figure 28). In this region, the fetch from the dominant wind directions is long (see Figure 29). The waves become smaller in the eastern parts of the Black Sea. Here, the wind speed decreases the same way as SWH and favours lower waves as in the western basin (not shown). The standard deviation ranges from 0.4 m in coastal areas to 0.7 m in the western part. The standard deviation is high on average where the mean significant wave heights are large. Regarding the maximum significant wave height, we see that the highest waves are found in the northern part of the central body of the Black Sea with values up to 9.5 m. Further local maxima can be detected in the south-west and in a small area in the south-east. The spatial distribution of the maximum wave height reflects single strong storm events causing high waves. The distribution of the 99th percentile of the significant wave height proves that most of the wave heights are below ~4 m with a maximum in the southwest of the Black Sea.

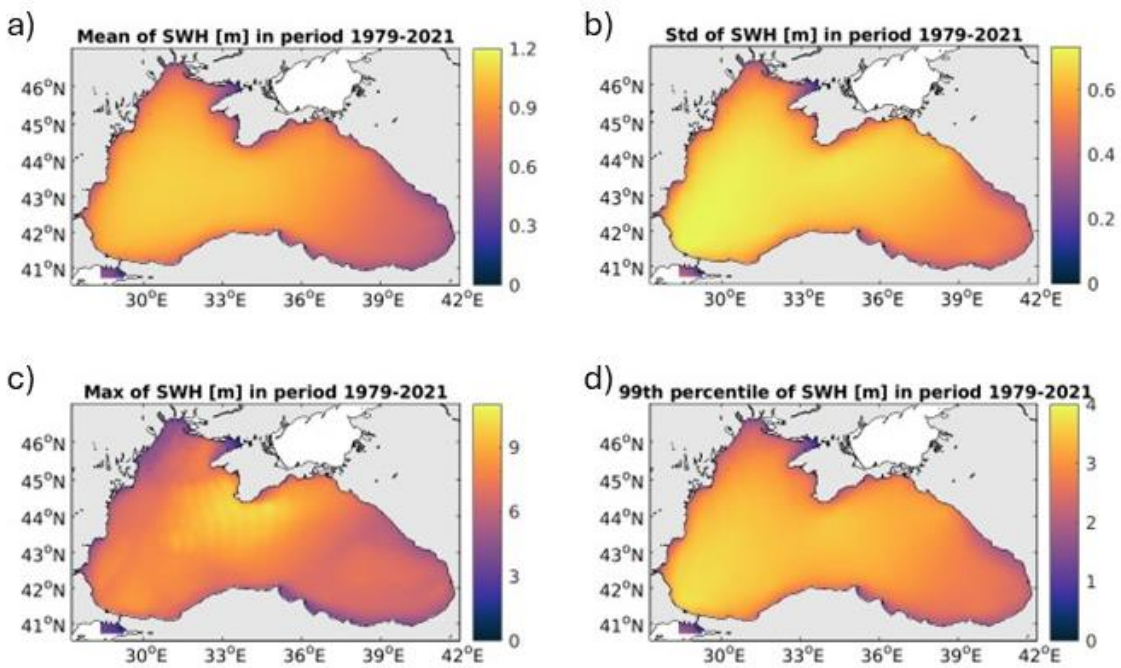

Figure 28: a) Average, b) standard deviation, c) maximum, and d) 99th percentile of significant wave height in WAM calculated over the period 1979-2021.

Beside the significant wave height, it is worth examining the distribution of mean wave directions (Figure 29). The dominant wave direction in the western part is to the west/south-west. In the eastern part, the waves on average go to the east/south-east. In the southern central Black Sea, the waves propagate on average to the south, whereas in the northern parts around the Crimean Peninsula the directions are variable.

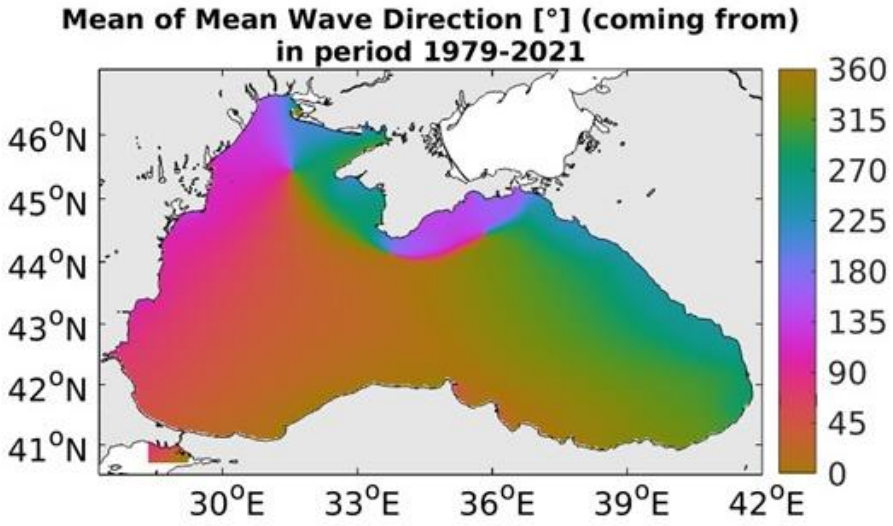

Figure 29: Average of the annual mean wave direction in WAM calculated over the period 1979-2021 shown in the "coming from" convention.

VIII.2 Histograms

The histograms in Figures 30 and 31 show the distribution of the annual mean of the significant wave height for the whole Black Sea and separately for the western and eastern basins (Figure 30) as well as the corresponding 99th percentiles (Figure 31). The annual mean for each of the years for the period 1979 – 2021 is distributed over the bins 2-5 in Figure 30a. Here the height of the individual columns in each of the bins indicate how many events or how many model grid points with proper values occur within the particular bins. The individual columns for each of the years are colour-coded, starting with dark green in 1979 and ends with bright yellow in 2021. The histogram demonstrates the frequency distribution per year and the variability over the years. While in the first bin 0 – 0.25 m there are very few values only, the number of occurrences increase continuously from the first bin to the fourth bin 0.75 – 1 m, which contains more than 50% of the annual means for all years. As an example (in Figure 30), some years (e.g., 2019) are unusual compared to the other columns. The values are much lower in the fifth bin compared to the fourth. Therefore, it is obvious that the highest wave heights are definitely lower in these years compared to all the other years.

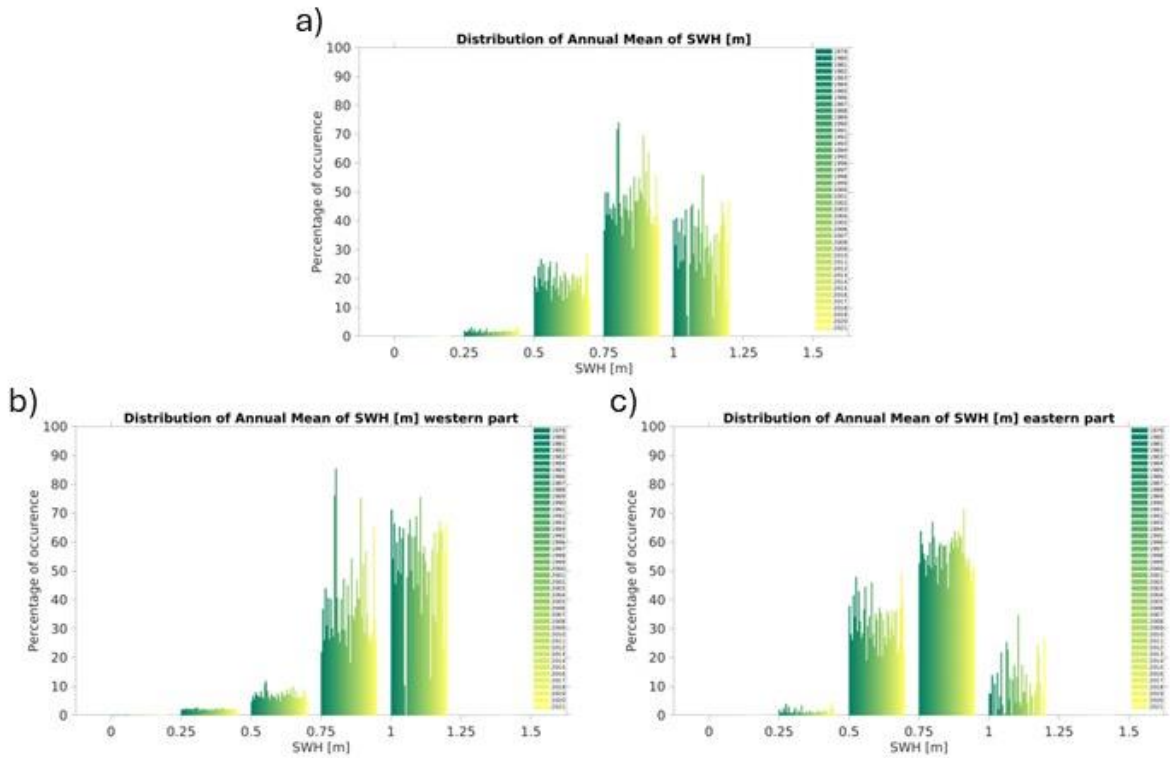

Figure 30: Histograms of the annual mean of the significant wave height (x-axis, in metres) for a) the whole Black Sea and additionally for the b) western and c) eastern part for the years 1979-2021.

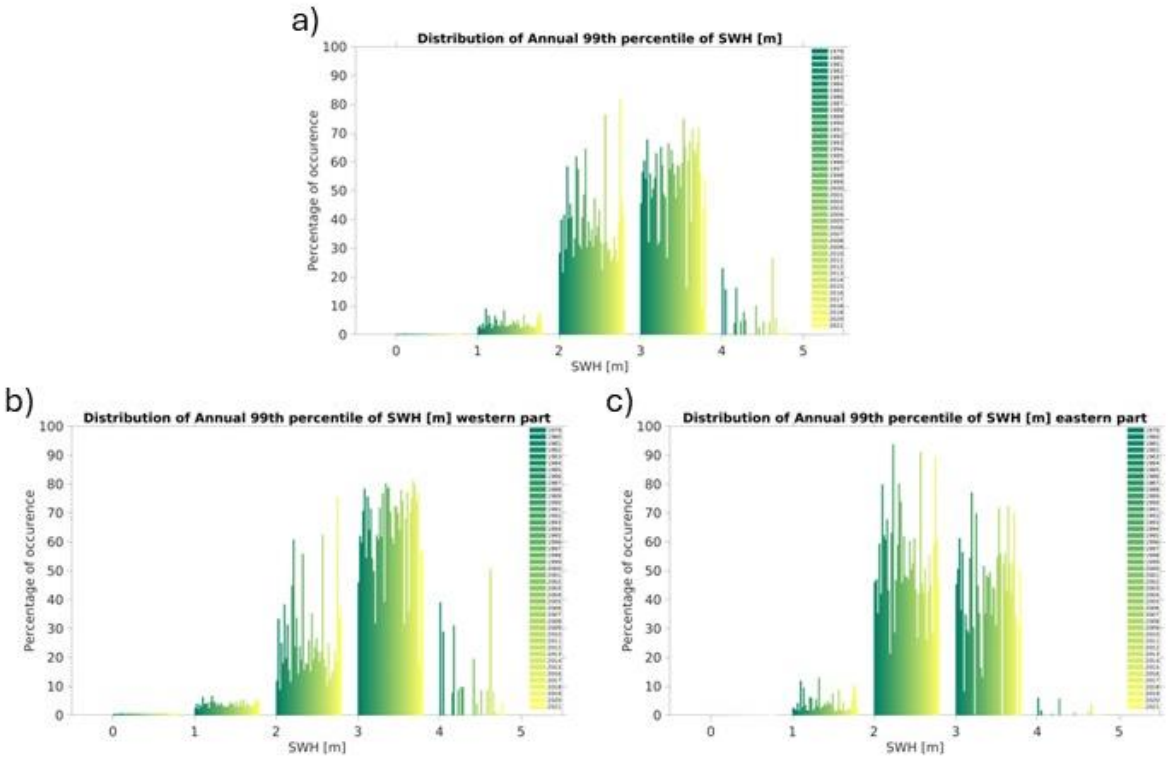

Figure 31: Histograms of the 99th percentile of the significant wave height (x-axis, in metres) for a) the whole Black Sea and additionally for the b) western and c) eastern part for the years 1979-2021.

The Figures 30b and c show the same histograms separated for the western and the eastern part of the Black Sea. The differences between the occurrences in the particular bins are quite significant. While for the western part, most of the occurrences can be found in the fifth bin in the eastern part of the Black Sea much less values are in this bin. Here most of the means occur in the fourth. That is a clear proof that the mean wave heights are significantly higher in the western part compared to the eastern part of the Black Sea.

The presentation of the values for the annual 99th percentile for the significant wave heights for the period 1979 – 2021 in Figure 31 supports the findings above, although the scatter of the occurrences for the individual years is higher than the one given in Figure 30. Furthermore, in this case the occurrences are distributed over five bins up to 5 m. In the last bin from 4 – 5 m the big storm in 2012 appears in Figure 31a, and in the plot for the western part of the Black Sea (Figure 31b).

VIII.3 Differences between western and eastern basin

With the significant wave height metrics in Figure 32, the different behaviour of the median, the mean and the maximum of the wave height as well as selected percentiles, has been investigated additionally for the western and the eastern part of the Black Sea. All values discussed here are averaged in space. The borderline between the western and the eastern part is defined as the longitude 34°W, as suggested

by the second empirical orthogonal functions (EOF) of the significant wave height in (not shown). The curves for the western and the eastern part of the Black Sea have been compared with that one for the complete area. It is obvious that the curves for the median and the mean of the significant wave height differ significantly. The wave heights in the western part are always higher than those generated for the eastern part. The maximum for the median in the western part is 0.88 m and 0.74 m for it the eastern part. For the mean of the significant wave height, the values are higher with 1.05 m for the western part and 0.90 m in the eastern part.

The maximum of the wave height for the period 1979 – 2021 in general is not much higher in the western part of the Black Sea. One of the few deviations of the eastern and western maximum is 1999 in which the eastern part has higher values.

Concerning the percentiles, it turns out that with increasing percentile, the differences between the western and eastern basins decrease. Finally, the differences between the averages of the significant wave heights in the western and eastern parts of the Black Sea can be explained by the different number of storms, the power of the storms, and the area that is covered by those.

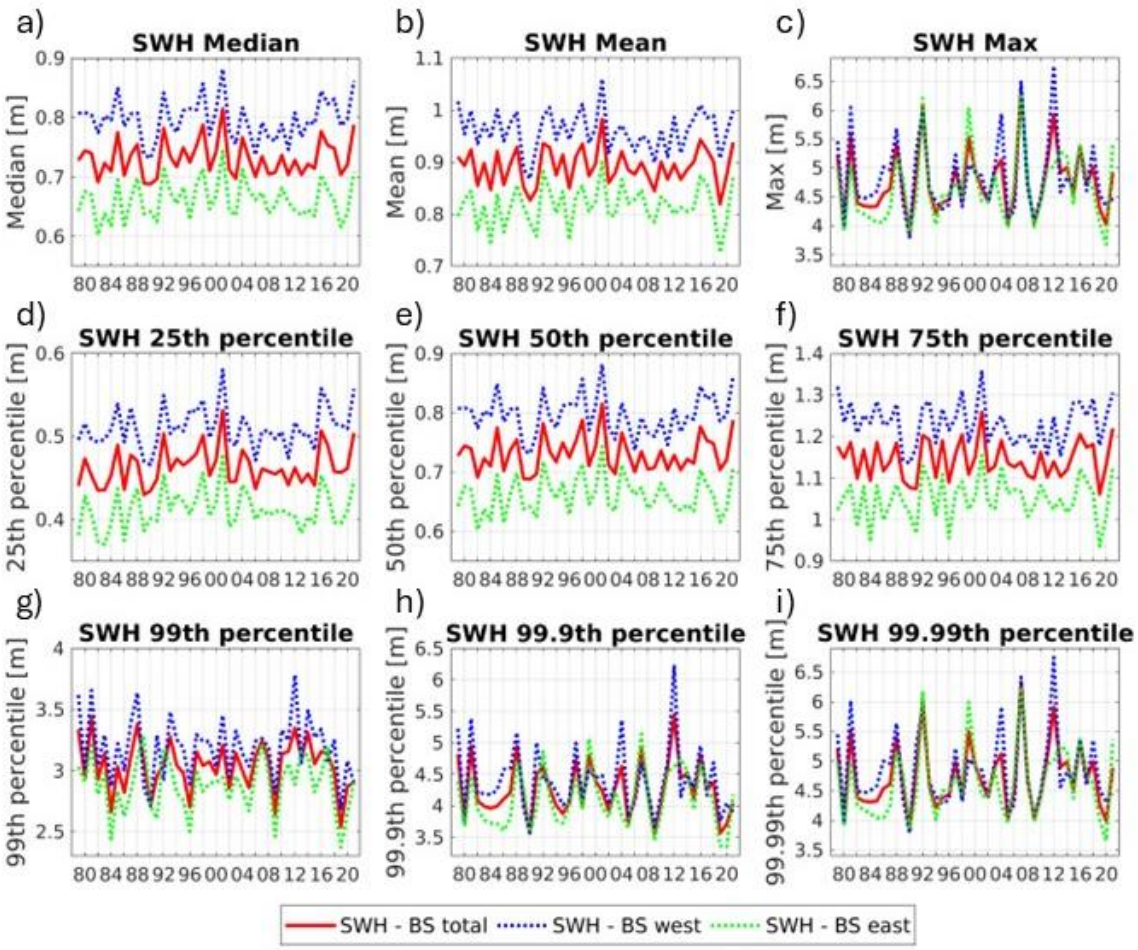

Figure 32: Yearly values of significant wave height metrics for the Black Sea derived from WAM model data shown for the whole basin (red line) as well as for the western (blue dashed line) and the eastern basin (green dashed line). The values are averaged in space. The western and eastern basin are separated at 34°W.

Investigating the maximum values in Figure 33 instead of the averaged values in Figure 32, the significant wave heights are higher of course and the differences for median, mean, maximum and the percentiles in the middle row of Figure 33 are less significant. The curves of the 25th, 75th and 99th percentiles are more similar now. The overall maximum of the significant wave height detected in the Black Sea during the period from 1979 to 2021 is 9.20 m in 2007. That is valid for both the wave height in the western and the eastern part since that event took place exactly on the borderline between the two different basins.

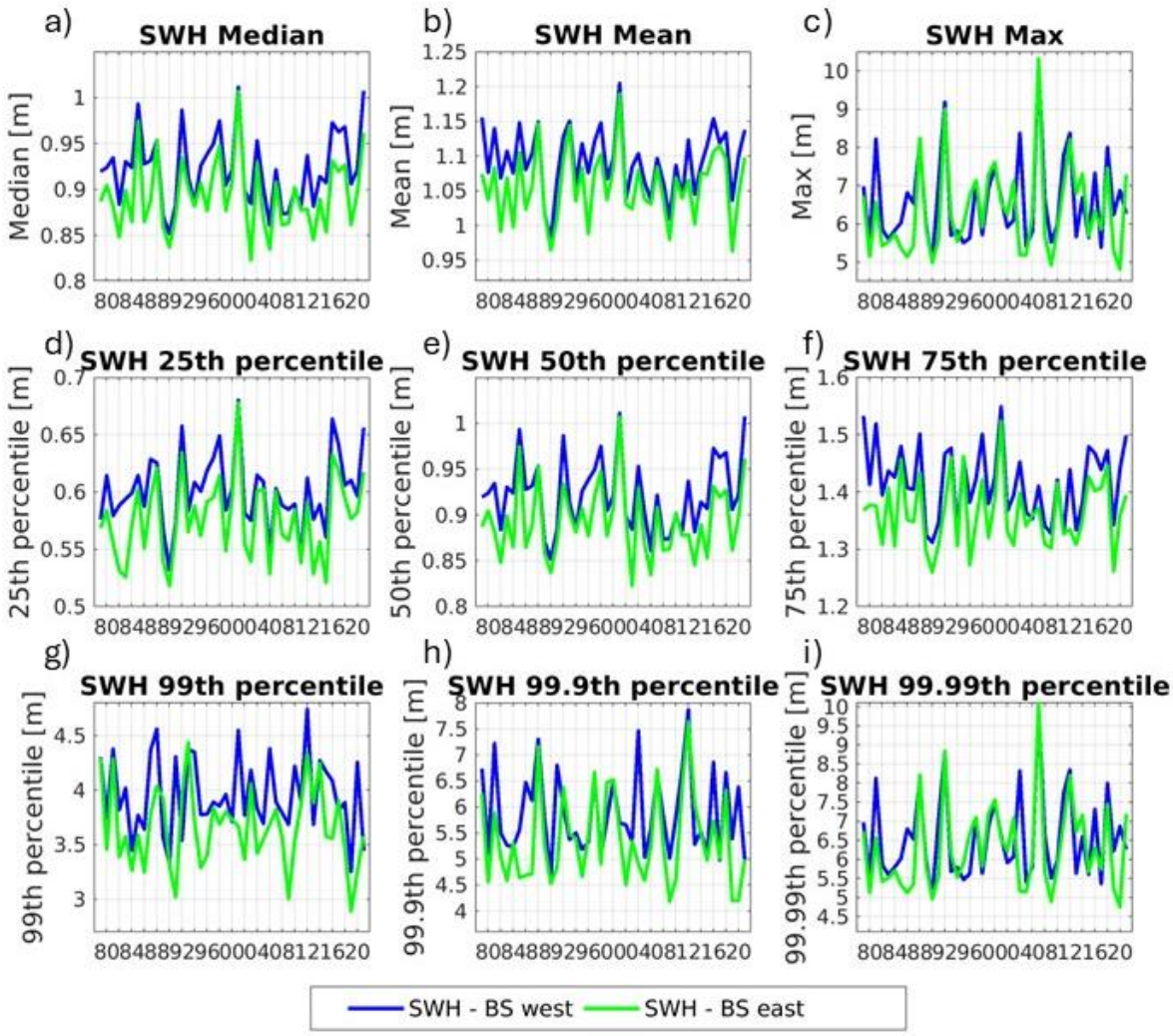

Figure 33: Yearly values of significant wave height metrics for the Black Sea derived from WAM model data shown for the western (blue line) and the eastern basin (green line). In contrast to Figure 32, the maximum values are shown.

VIII.4 Climatology

Figures 34 and 35 show the monthly climatology for significant wave height and Tm02 period, respectively, from the dataset *cmems_mod_blk_wav_my_2.5km-climatology_PT1M-m*.

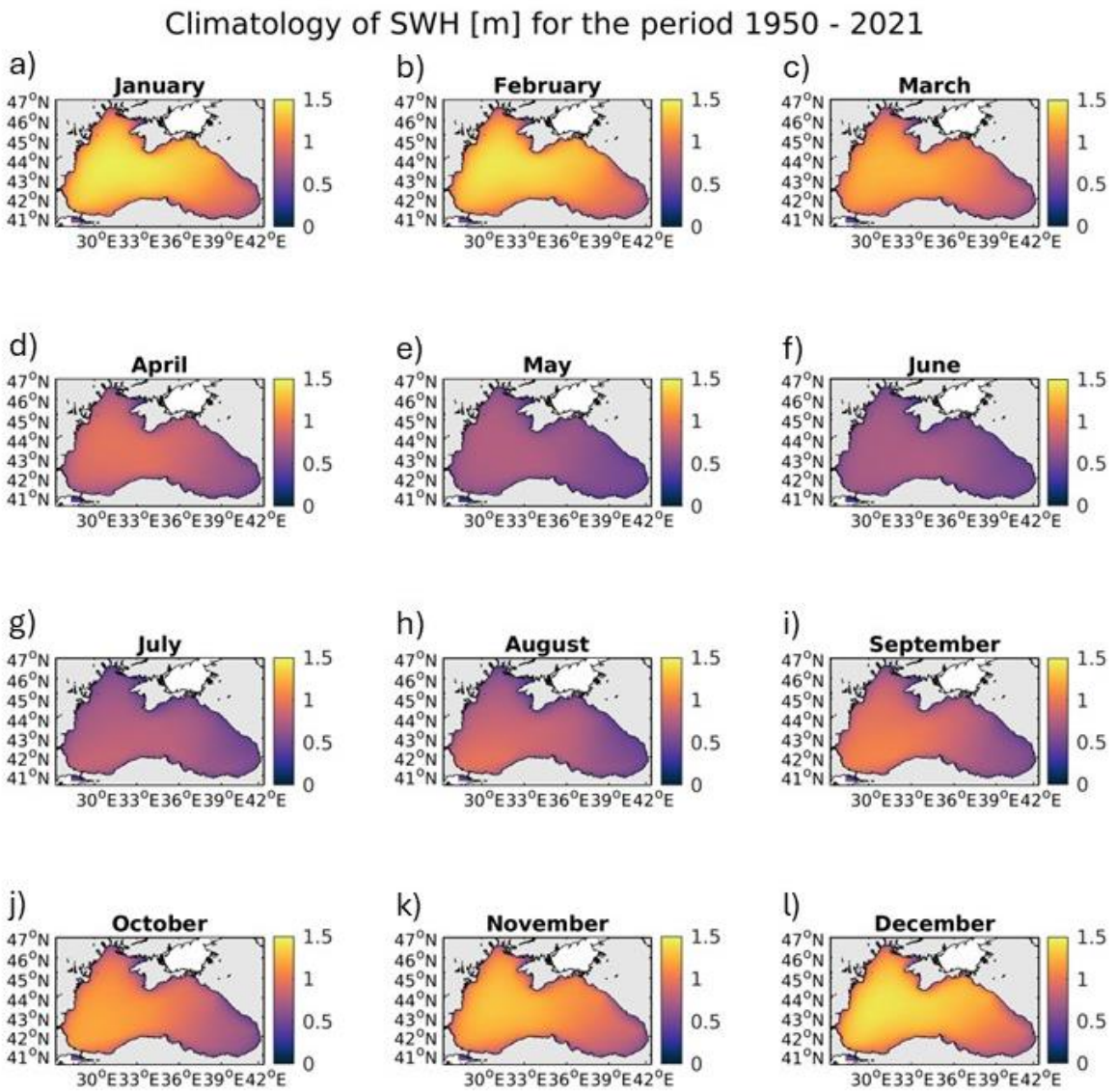

Figure 34: Monthly climatology (from a) January to l) December) of the MY product for significant wave height based on the period 1950 to 2021.

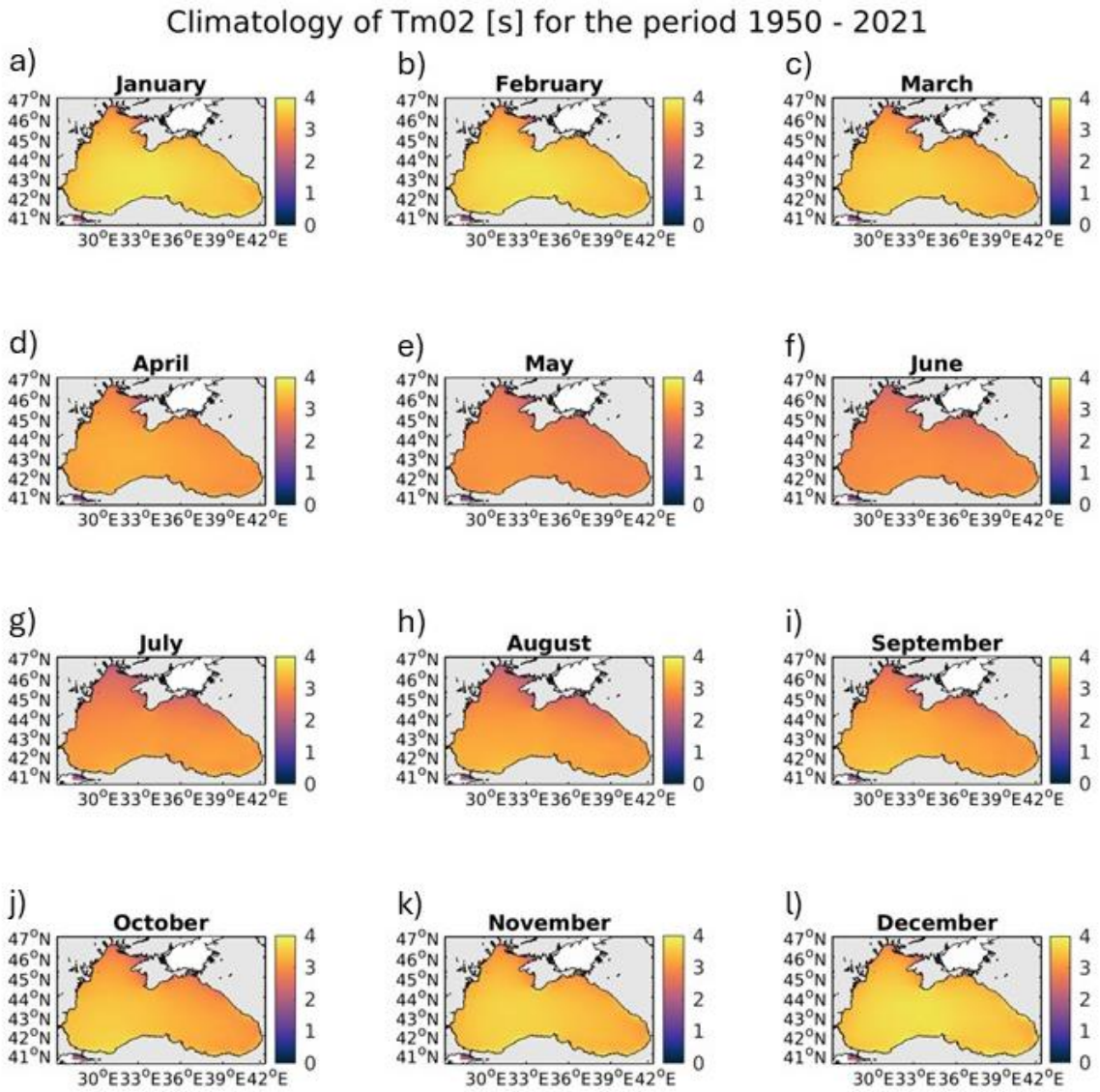

Figure 35: Monthly climatology (from a) January to l) December) of the MY product for Tm02 period based on the period 1950 to 2021.